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Tri-Attention boosted scalable EfficientNet for skin lesion classification via Triple-Stage Gain Ratioed Averaging

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ABSTRACT

Skin lesions appear in different abnormal forms, and skin diseases often disrupt both the structure and functionality of the skin. Early diagnosis and automated prediction are crucial, particularly since skin cancer is characterized by uncontrolled cellular proliferation. Despite progress in this area, a major research challenge lies in accurately locating the dominant regions of dispersion after lesion detection. To address this, we present an innovative framework based on Transfer Learning, incorporating multiple variants of Customized EfficientNet. Our approach integrates EfficientNet versions with three distinct attention mechanisms — Channel Attention, Soft Attention, and Squeeze-Excitation Attention — leading to the development of a scalable EfficientNet architecture. To further enhance performance, we introduce a specialized Triple-Stage Ensemble Learning strategy, termed Gain Ratioed Averaging (GRA), which effectively tackles the problem of optimal weight allocation for individual models within the ensemble. The central aim of GRA is to achieve efficient aggregation by determining the most suitable weights, and its extension, Triple-Stage GRA (TS-GRA), allows for broader application across three stages. Experimental results on the HAM10000 dataset validate our method, achieving a remarkable accuracy of 94.57% with TS-GRA, surpassing state-of-the-art approaches. Moreover, to improve interpretability and reveal the regions influencing the model's decisions, we employ Gradient-weighted Class Activation Mapping (Grad-CAM). This visualization technique strengthens explainability by clearly highlighting the critical areas relevant to diagnosis.

1. Introduction

The skin's health and function can be compromised by a wide array of issues, broadly classified as skin diseases. These conditions range from common and minor, like acne, to serious illnesses such as skin cancer. A key characteristic of these diseases is the development of skin lesions, which are abnormal changes in the skin's appearance. However, the full scope of a skin disease often extends beyond visible lesions to include various other symptoms. The origins of these lesions are highly diverse, stemming from causes like infections, allergic or inflammatory reactions, genetic factors, autoimmune diseases, environmental exposures, injuries, vascular issues, and malignancies such as cancer. Each cause is linked to a unique set of features and a distinct disease process. Based on their potential for harm, skin lesions are typically categorized into two groups. The first is benign lesions, which are non-cancerous and generally pose no health risk. Common examples include moles, skin tags, warts, seborrheic keratoses, and hemangiomas. In contrast, malignant lesions are cancerous growths capable of spreading to other parts of the body (metastasizing). The most prevalent malignant types are basal cell carcinoma, squamous cell carcinoma, and melanoma.

Early recognition of skin irregularities is therefore critical. However, many patients remain undiagnosed due to the complexity and cost of medical evaluations. Dermatoscopy — also known as dermoscopy or epiluminescence microscopy — is a non-invasive diagnostic technique in dermatology. By using a handheld magnification device equipped with light, it allows detailed examination of skin lesions. While this tool improves early detection compared to traditional methods, its diagnostic accuracy heavily depends on clinician expertise, leaving room for human error.

In contrast, automated systems powered by artificial intelligence (AI), particularly machine learning (ML) and deep learning (DL), offer promising avenues for lesion detection. By analyzing medical images, these methods accelerate early diagnosis, raise disease awareness, and improve treatment outcomes. Despite progress, challenges persist: model training often favors classes with abundant data; transfer

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learning (TL) models pretrained on ImageNet may fail to capture task-specific features unless carefully fine-tuned; and ensemble strategies struggle with optimal weight allocation across models. Additionally, many studies overlook interpretability, and biases may arise when the same dataset is used for both validation and testing.

Previous work has frequently applied convolutional neural networks (CNNs) and ensemble learning (EL) to overcome individual model limitations [1]. Yet, the absence of systematic weight optimization across models remains a critical obstacle. Traditional ensemble strategies — such as majority voting, uniform softmax averaging, or arbitrarily weighted predictions — rarely account for the differing contributions of individual models, limiting their effectiveness.

To address these issues, our study introduces a novel framework that directly responds to the following research questions:

RQ1: *Given the variability in image resolutions, how can a scalable transfer learning architecture be designed?*

– TL models are often tailored to specific input sizes. Combining variants that process multiple resolutions into a unified scalable architecture can improve performance.

RQ2: *How can attention mechanisms be leveraged to emphasize the most critical image regions?*

– Redundant or irrelevant portions of an image may degrade classification accuracy, whereas certain regions are crucial for correct diagnosis. Attention-based methods can help models focus on these significant regions.

RQ3: *Is a single model sufficient, or should ensemble strategies be employed for robustness?*

– Since no single model consistently excels across all data distributions, reliance on a single architecture is limiting. Ensemble learning, enhanced with explainable AI (XAI), can provide stronger generalization.

RQ4: *What are the drawbacks of conventional ensemble methods, and why is a new approach needed?*

– Existing methods do not guarantee optimal weight distribution across predictors. A more principled strategy is required to determine the ideal contribution of each model.

Motivated by these challenges, this research makes the following key contributions:

1. We propose a scalable transfer learning framework using all EfficientNet variants, enabling the processing of images across multiple resolutions within a unified architecture.
2. To improve feature focus, three attention mechanisms — Channel Attention (CA), Soft Attention (SA), and Squeeze-Excitation Attention (SEA) — are incorporated into customized Scalable EfficientNet (SEN) models, highlighting the most relevant aspects of the input data.
3. We introduce a novel ensemble strategy, Gain Ratioed Averaging (GRA), which dynamically assigns optimal weights to EfficientNet variants. Unlike conventional weight assignment methods, GRA operates across multiple stages, yielding more effective aggregation.
4. To ensure interpretability, we employ Gradient-weighted Class Activation Mapping (Grad-CAM), enabling visualization of the regions most influential to the model's decision-making and thereby improving transparency.

2. Literature review

Skin lesion classification has been widely investigated, with numerous researchers contributing methods that explore both custom architectures and transfer learning techniques. This section highlights key approaches, their contributions, and their limitations. Studies such as [2–4] employed customized CNNs, while others [5–17] leveraged transfer learning (TL) and attention mechanisms to refine performance.

Sevli [2] introduced a CNN model deployed via a web-based REST API for skin lesion diagnosis. The model achieved 91.51% accuracy in dermatologist-led evaluations but suffered from limited feature discrimination due to the absence of attention mechanisms. Similarly, Shetty et al. [3] applied CNNs to detect skin cancer with 94% accuracy, though their reliance on a small augmented subset (200 images per class) restricted generalizability. Saarela and Georgieva [4] adopted a Bayesian inference-based method to improve interpretability, yet their accuracy of 80% on test data was comparatively modest.

Sun et al. [18] enhanced classification by integrating metadata and advanced augmentation strategies, achieving 88.7% accuracy with a single model and 89.5% with an embedding approach. However, their augmentation procedure lacked transparency. Hoang et al. [19] employed a segmentation-driven strategy with a lightweight ShuffleNet, yielding accuracies between 84.8% and 86.33%. Nie et al. [20] combined CNNs with transformers and focal loss to capture both low-level and high-level features, attaining 89.48% accuracy. Despite promising results, their reliance on deeper feature extraction posed limitations. Efat et al. [5] proposed a new ensemble technique that used gini index reversely though the best accuracy was not up to the mark.

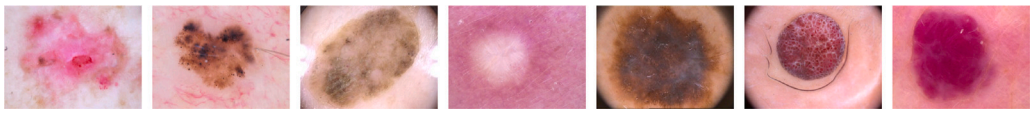
Rahman et al. [6] proposed an ensemble of five TL-based models using weighted averaging, reaching 88% accuracy. Nevertheless, their direct use of pre-trained weights restricted dataset adaptability. Mahbod et al. [7] examined input resolution in TL and introduced a multi-CNN fusion strategy that produced 86.2% accuracy, albeit at the cost of increased model complexity. Efat [8] proposed the optimal weighting approach by which the best performance can be guaranteed. Harangi et al. [9] combined binary and multiclass classification outcomes, improving accuracy to 93.46%. However, the reasoning for integrating binary results was not fully justified. Wang et al. [10] developed a dual-stream framework fusing DenseNet-121 and VGG-16, achieving 91.24% accuracy, but with limited fine-tuning of pre-trained models.

Popescu et al. [11] applied collective intelligence techniques with TL, reaching 86.71% accuracy on validation data, though independent testing results were not reported. Khan et al. [12] integrated ResNet50 with a feature pyramid for segmentation, followed by a deep CNN classifier, achieving 86.5% accuracy. Their method, however, overlooked segmentation mask information from HAM10000. Nigar et al. [13] developed an interpretable model using ResNet-18 with LIME, reaching 94.47% accuracy. While effective, the method relied on a single pre-trained model and a limited dataset. Gouda et al. [14] applied ESRGAN-based image enhancement before CNN classification, achieving 83.2% accuracy. Yet, imbalanced datasets remained unaddressed in their design.

Attention-based models have also been widely studied. Datta et al. [15] incorporated Soft Attention for improved classification, achieving 93.4% accuracy across two datasets, though their method did not optimize attention across color channels. Similarly, Nguyen et al. [16] integrated Soft Attention with InceptionResNetV2 and MobileNetV3Large, reporting accuracies of 90% and 86% respectively, but without comparison to other attention mechanisms. Efat et al. [17] proposed multiple attention mechanisms to be used in a single framework. Shafin et al. [21] used attentions sequentially to obtain the best performance.

The studies [22–30] explored multiple attention modules but with only one model with no ensemble approach.

Drawing on these prior works, our study addresses several unresolved gaps. First, we utilize the complete dataset, mitigating class imbalance through enhanced training strategies while maintaining an independent test set. Second, we integrate Attention Techniques (AT) with TL to prioritize clinically significant regions. Third, we refine TL models through a Scale-Enhanced Network (SEN) that reduces dependence on ImageNet pre-training. Finally, to handle variability in input resolutions, all EfficientNet variants are incorporated into a unified scalable framework, ensuring robust and adaptable classification.



(a) Actinic Keratosis (AK) (b) Basal Cell Carcinoma (BCC) (c) Benign Keratosis (BKL) (d) Dermatofibroma (DF) (e) Melanoma (MEL) (f) Nevus (NV) (g) Vascular Lesions (VASC)

Fig. 1. Representative samples from each lesion class in the HAM10000 dataset.

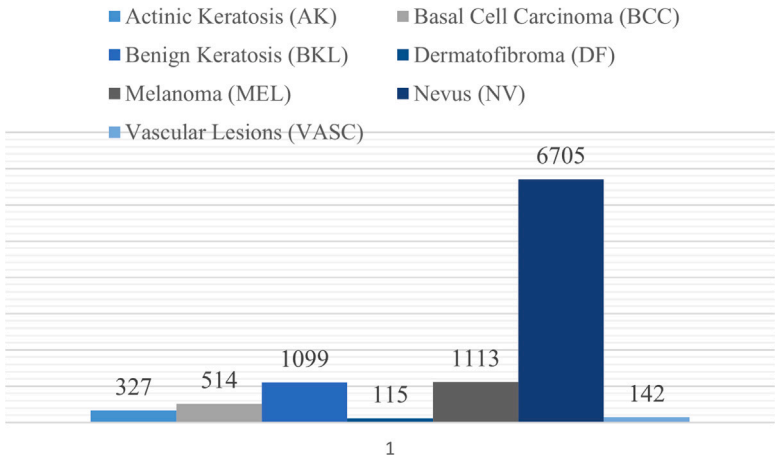


Fig. 2. Distribution of samples across classes, showing dataset imbalance.

Table 1
Overview of the HAM10000 dataset.

Images	Format	Number of classes	Source
10015	JPG	7	Harvard Dataverse

3. Dataset description

This study makes use of the Human Against Machine (HAM10000) dataset, which is publicly available through the Harvard Dataverse repository [31]. The dataset was carefully curated to include a wide range of dermatoscopic skin lesion images. It contains 10,015 images in JPEG format, categorized into seven distinct classes: Actinic Keratosis, Basal Cell Carcinoma, Benign Keratosis, Melanoma, Dermatofibroma, Nevus, and Vascular lesions. Among these categories, Melanoma, Actinic Keratosis, and Basal Cell Carcinoma are malignant in nature, whereas Nevus, Benign Keratosis, and Dermatofibroma are benign. Vascular lesions may also present malignant characteristics in certain cases. A concise depiction of the dataset is presented in Table 1.

Representative examples from each lesion category are shown in Fig. 1, with one image provided per class. Furthermore, the class imbalance inherent in the dataset is visualized in Fig. 2, emphasizing the uneven distribution of samples across categories.

4. Research methodology

The proposed methodology follows a sequential pipeline beginning with dataset preparation, followed by essential preprocessing and augmentation. The dataset is partitioned into training, validation, and testing subsets. To mitigate the class imbalance problem, augmentation strategies are applied exclusively to the training set, while validation and test sets remain untouched to ensure unbiased evaluation. Next, Scalable EfficientNet (SEN) architectures are developed and trained using the training data, with validation serving for hyperparameter adjustment. Model performance is finally assessed on the test set. Predictions from each SEN variant are integrated using the Gain Ratioed Averaging (GRA) technique, which is evaluated across three distinct stages. To enhance interpretability, Grad-CAM is applied to visualize the discriminative regions influencing model predictions. The overall workflow is illustrated in Fig. 3.

4.1. Data preprocessing and augmentation

The preprocessing step begins by organizing samples according to their lesion IDs, ensuring that no duplicate samples appear across different subsets. The dataset is divided into training, validation, and test splits, with augmentation applied solely to the training set to preserve independence of the evaluation sets. Additional redundancy is introduced in the training subset to increase generalization ability. The augmentation process aims to create roughly 8000 samples per class, thereby alleviating imbalance.

To achieve this, TensorFlow's `ImageDataGenerator` is used. Prior to augmentation, image contrast is enhanced to improve lesion visibility. The augmentation pipeline includes random rotations (up to 180°), horizontal and vertical flipping, width and height shifts (10%), and zoom operations (within 10%). Any gaps introduced are filled using nearest-neighbor interpolation to maintain visual coherence. This strategy significantly increases intra-class diversity and strengthens model robustness. The augmented dataset distribution results in: AK (7854), BCC (7965), BKL (7944), DF (7377), MEL (7932), NV (8004), and VASC (7706). Example augmented images are shown in Fig. 4.

4.2. Scalable EfficientNet (SEN) with Tri-Attention (TA)

Transfer Learning (TL) is leveraged using EfficientNet (EN) variants as base feature extractors. Two families of EN are employed: EfficientNetv1 (ENv1) with models b0–b7 and EfficientNetv2 (ENv2) with b0v2–b3v2, ENv2S, ENv2M, and ENv2L. These networks differ in input image resolutions, hence they are referred to collectively as Scalable EfficientNet (SEN). Table 2 lists input sizes for each model.

Parameter Efficiency vs. Accuracy Trade-Off: To balance model complexity and predictive performance, we utilize different input image sizes for each EfficientNet variant, following the recommendations

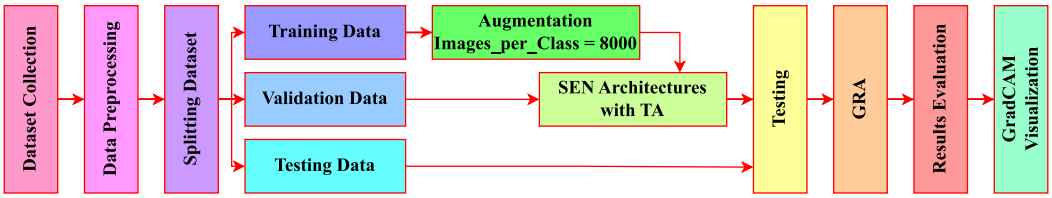


Fig. 3. Workflow representation of the proposed methodology.

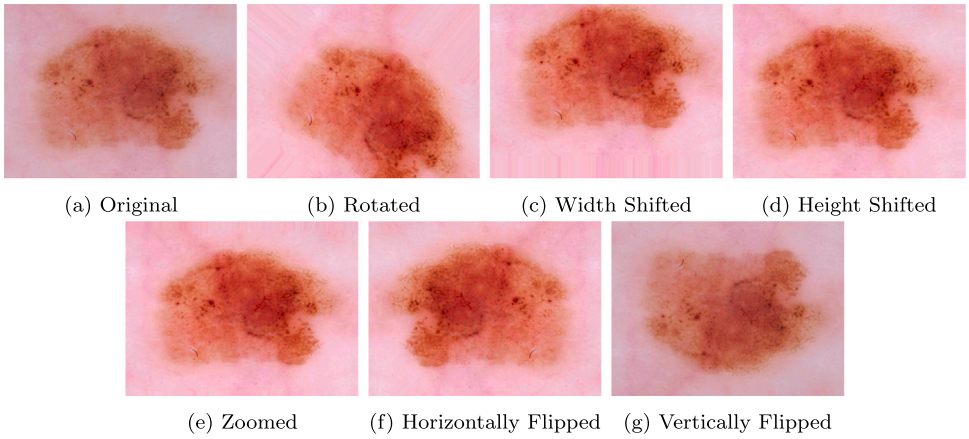


Fig. 4. Examples of original and augmented images.

Table 2
Input resolutions for SEN architectures.

Model	b0	b1	b2	b3	b4	b5	b6	b7
Size	224	240	260	300	300	300	300	300

Model	b0v2	b1v2	b2v2	b3v2	ENv2S	ENv2M	ENv2L
Size	224	240	260	300	224	240	300

of the original authors. This approach allows smaller models to operate with fewer parameters while preserving near-optimal accuracy, and larger variants to exploit higher-resolution inputs for improved discriminative power. By combining these models in a TS-GRA ensemble, we achieve a favorable trade-off: the ensemble benefits from diverse and accurate feature representations without excessively increasing the total number of parameters, maintaining both efficiency and high performance.

Since pre-trained weights are not optimized for the HAM10000 dataset, fine-tuning is performed using four custom CNN backbones: a plain CNN (CCNN), CNN with Channel Attention (CACNN), CNN with Squeeze-Excitation Attention (SEACNN), and CNN with Soft Attention (SACNN). The complete SEN architecture is depicted in Fig. 5.

4.2.1. Scalable EfficientNet with basic fine-tuning blocks

The integration of pre-trained EfficientNet (EN) models with our customized CNN variants (CCNN, CACNN, SEACNN, or SACNN) begins by importing the EN architectures available in the `TensorFlow` library. Each model is initialized with the designated input shape, which reformats the output into a four-dimensional tensor: `[None, height, width, channels]`. This modification ensures compatibility with our framework, since EfficientNet natively produces a two-dimensional output that does not align with the required input dimensions of our architecture.

Following this adjustment, the fine-tuning process is systematically applied to adapt the pre-trained weights to our specific dataset. This involves unfreezing selected layers of the EfficientNet backbone while retaining lower-level features, allowing the model to learn domain-specific representations. Finally, the predictions generated by each individual model are retained for subsequent evaluation and ensemble integration.

4.2.2. Design of fine-tuning blocks with customized CNN and tri-attention (TA)

The Customized CNN (CCNN) framework is constructed using two convolutional blocks, each containing four Conv2D layers with kernel sizes of 7×7, 5×5, 3×3, and 1×1, followed by corresponding BatchNormalization layers. A MaxPooling2D operation is applied at the end of each block to reduce spatial dimensionality. The first block employs 128 filters, while the second uses 256 filters. To alleviate vanishing gradient issues and ensure efficient feature propagation, all convolutional layers adopt the ReLU activation function.

In the Channel Attention CNN (CACNN), a Channel Attention (CA) module is incorporated within each convolutional block of the CCNN. Specifically, the CA module is positioned after each Conv2D and its associated BatchNormalization layer. This intermediate placement allows channel-wise refinement of features during convolution, amplifying informative signals while suppressing less discriminative ones.

The Squeeze-and-Excitation CNN (SEACNN) integrates the SEA module after the completion of each convolutional block. By recalibrating the importance of channels globally, the SEA layer enables adaptive emphasis on meaningful features at a higher level of abstraction, thereby improving the capture of hierarchical patterns.

Similarly, the Soft Attention CNN (SACNN) inserts a Soft Attention (SA) module after each convolutional block, following the same principle as SEACNN. However, due to the substantial parameter overhead of SA, it is not applied after every individual Conv2D layer. Instead, SA is strategically placed after the blocks to balance computational cost with improved focus on spatially significant regions.

The feature maps generated by the final max-pooling layer are flattened into a one-dimensional vector and processed by a fully connected block. This block comprises three sequential dense layers with 1024, 512, and 7 units. The first two layers utilize the ReLU activation function, while the final layer uses a softmax activation to generate

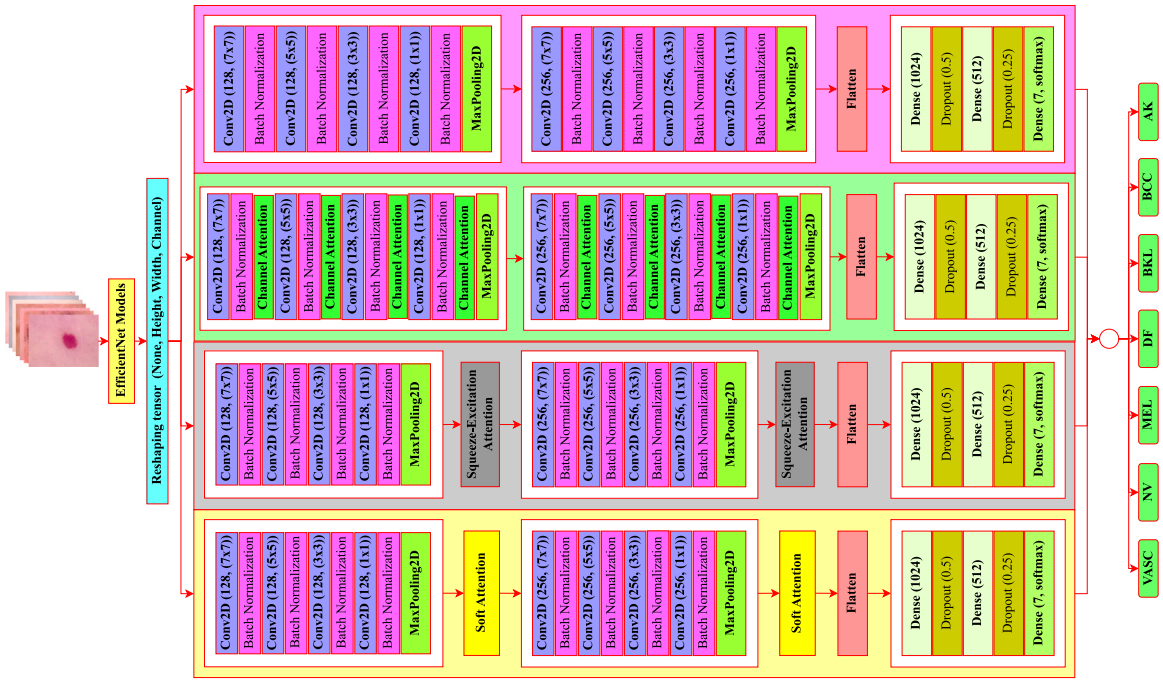


Fig. 5. Scalable EfficientNet (SEN) framework integrating Tri-Attention.

a probability distribution across the seven output classes. To mitigate overfitting, dropout regularization is applied after the first two dense layers with rates of 0.5 and 0.25, respectively.

4.2.3. Feature extraction process

For feature extraction, the pre-trained EfficientNet-B0 architecture is utilized as a backbone, with its final classification layers removed (include_top = False). The output from this backbone is subjected to a global average pooling operation (pooling = 'avg'), which aggregates spatial information into a condensed feature vector. This vector is then reshaped into a format (None, height, width, channels) to be processed by a series of convolutional layers. These layers, with kernel sizes of 7×7, 5×5, 3×3, and 1×1, each employ a ReLU activation function and are followed by batch normalization to stabilize the learning process and introduce non-linearity. To gradually decrease spatial dimensions and improve feature discrimination, max-pooling layers are strategically applied. The resulting feature maps are flattened and fed into a stack of fully connected layers with ReLU activations, culminating in a final dense layer that uses a softmax activation function to output probabilities for each class.

Fig. 6 depicts the activation of features at multiple levels of a customized EfficientNet-B0 transfer learning model, with each row representing a different architectural layer.

– **Input Layer (`input_1`):** Shows the preprocessed input image after normalization.

– **Zero Padding (`zero_padding2d`):** Illustrates feature maps after zero-padding is applied to maintain spatial dimensions before convolution.

– **Convolution (`conv2d`):** Visualizes activations from 64 convolutional filters, highlighting basic textures and edges.

– **Batch Normalization (`batch_normalization`):** Displays standardized activations that promote training stability.

– **ReLU Activation (`activation`):** Reveals feature maps after rectified linear unit non-linearity is introduced.

– **Max Pooling (`max_pooling2d`):** Presents down-sampled feature maps that retain the most salient information.

– **Concatenation (`concatenate`):** Demonstrates the merging of feature maps from multiple pathways.

– **Dense Layer (`dense`):** Shows the transformation into high-level vector representations before classification.

– **Output Layer (`dense_1`):** Depicts the final softmax output representing class probability scores.

Each sub-figure visualizes a maximum of five filters per layer using the viridis color scheme for clarity. These visualizations trace the hierarchical refinement of features from raw pixels to class-specific predictions, elucidating the model's internal feature extraction process. Although based on a single input sample and a limited set of layers, the figure exemplifies how complex feature representations are constructed, ultimately contributing to the model's classification performance.

4.3. Tri-Attention (TA)

To improve feature representation, our framework incorporates three complementary attention mechanisms. These modules are designed to emphasize the most informative features while reducing the impact of less significant ones.

4.3.1. Channel attention (CA)

The channel attention mechanism refines feature maps by assigning adaptive weights to each channel. This process leverages mean and variance statistics to capture channel dependencies, enabling the model to highlight discriminative channels while suppressing irrelevant ones [32].

$$\mathbf{w_c} = \sigma\left(\mathbf{W_2}\,\delta\left(\mathbf{W_1}\mathbf{x}\right)\right) \tag{1}$$

$$\mathbf{y_c} = \mathbf{w_c} \odot \mathbf{x} \tag{2}$$

Where,

– $\mathbf{x}$ denotes the input feature maps of size $C \times H \times W$,

– $\mathbf{W_1}$ and $\mathbf{W_2}$ are learnable weight matrices,

– δ and σ represent the ReLU and sigmoid activation functions, respectively,

– $\mathbf{w_c}$ corresponds to the learned channel attention weights, and $\odot$ indicates element-wise multiplication [32].

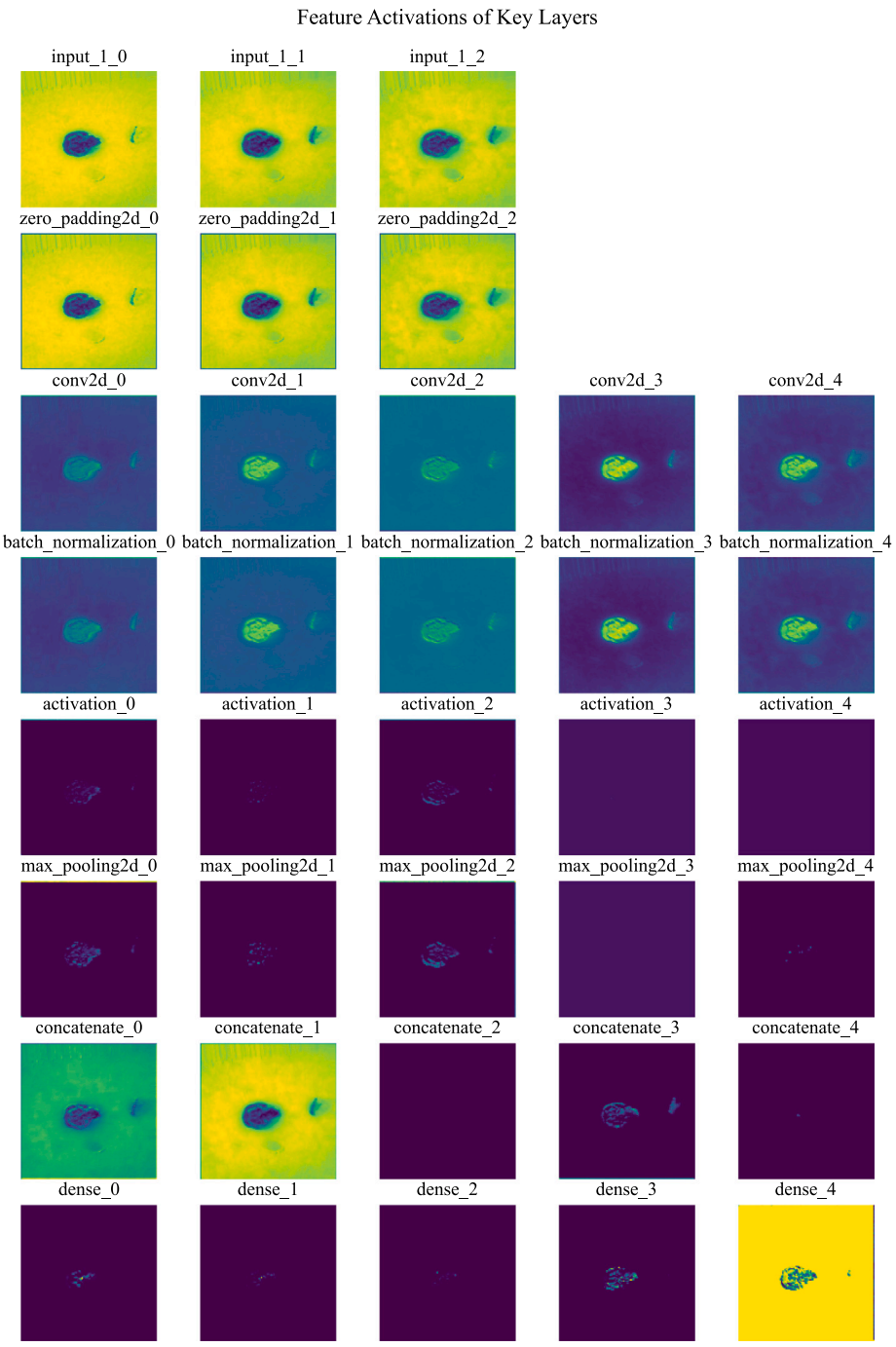


Fig. 6. Feature extraction at successive activation layers using a single image as an example.

4.3.2. Squeeze-Excitation Attention (SEA)

The SEA module enhances representational capacity by first compressing spatial information through global average pooling, followed by learning inter-channel dependencies. This mechanism recalibrates feature responses, enabling the model to adaptively reweight channels based on their importance [33].

Given input feature maps $\mathbf{x}$ of size $C \times H \times W$:

$$\mathbf{z} = \text{GlobalAvgPooling}(\mathbf{x}) \tag{3}$$

$$\mathbf{s} = \text{ReLU}(\mathbf{W}_2\, \sigma(\mathbf{W}_1 \mathbf{z})) \tag{4}$$

$$\mathbf{y} = \mathbf{s} \odot \mathbf{x} \tag{5}$$

4.3.3. Soft attention (SA)

The soft attention mechanism assigns weights to individual input elements, allowing the model to selectively focus on spatial regions with higher discriminative value. The attention score for the ith element is computed as follows [15]:

$$a_i = \frac{\exp(e_i)}{\sum_{j=1}^{T} \exp(e_j)} \tag{6}$$

Where,

- a_i is the normalized attention weight for the ith element,
- T represents the sequence length, and
- e_i denotes the unnormalized score for the ith element [15].

4.4. Gain Ratioed Averaging (GRA)

We propose a novel ensemble learning strategy, termed *Gain Ratioed Averaging (GRA)*, designed to assign optimal weights to predictions from individual classifiers. The method combines entropy, information gain (IG), and gain ratio (GR) to derive these weights and aggregate predictions through weighted averaging. The procedure can be described in the following steps.

Step 1: Entropy computation. The process begins by evaluating the entropy of classifier predictions. Correct predictions are labeled as '1', while incorrect ones are labeled as '0'. Entropy quantifies the uncertainty of this binary random variable X:

$$H(X) = -\sum_{i=1}^{n=2} p(x_i)\log_2(p(x_i)), \tag{7}$$

where
- $H(X)$ is the entropy of the random variable X,
- $p(x_i)$ denotes the probability of outcome x_i, and
- $n = 2$ since there are two possible outcomes.

Entropy reaches its maximum when both outcomes are equally probable and decreases when the distribution becomes more skewed.

Step 2: Information gain. Information gain measures the reduction in entropy after a split:

$$IG = \text{Entropy before split} - \sum_{i=1}^{n} \frac{|S_i|}{|S|} \cdot Entropy(S_i), \tag{8}$$

where
- S is the original dataset, and S_i represents each subset,
- $|S|$ and $|S_i|$ denote the sizes of S and subset S_i, respectively,
- $Entropy(S_i)$ is the entropy of subset S_i.

Step 3: Gain ratio. To avoid bias toward attributes with many distinct values, gain ratio (GR) is employed:

$$GR = \left(\frac{IG}{SI}\right)^{\sigma} \tag{9}$$

where
- IG is the information gain, and
- SI is the split information, defined as:

$$SI = -\sum_{i=1}^{n} \frac{|S_i|}{|S|}\log_2\left(\frac{|S_i|}{|S|}\right). \tag{10}$$

Role of σ: The parameter σ controls the influence of each classifier within the ensemble. By adjusting this parameter, classifiers with higher accuracy and stronger GR values are emphasized, ensuring that consistently superior models dominate the ensemble prediction. This prevents weaker classifiers from diluting performance while enhancing robustness across multiple stages.

Step 4: Weight assignment. Once GR values are computed, weights are derived as the normalized contribution of each classifier:

$$weight_i = \frac{GR_i}{\sum_{i=1}^{n} GR_i}, \tag{11}$$

where n is the number of classifiers.

Step 5: Weighted averaging. The final ensemble prediction is obtained by combining individual outputs according to their weights. Let K be the number of classifiers. Each classifier m produces predictions $Q_m = [q_{m1}, q_{m2}, \ldots, q_{mk}]$, where k is the number of instances. The ensemble prediction for instance n is computed as:

$$F_n = \sum_{m=1}^{K} v_m \cdot q_{mn}, \tag{12}$$

where
- F_n is the final ensemble output for instance n,
- v_m is the weight for classifier m (derived from GR), and
- q_{mn} is the prediction of classifier m for instance n.

This weighted averaging ensures that classifiers with higher gain ratios exert greater influence in the ensemble, thereby improving the overall predictive performance. Fig. 7 illustrates the GRA process for a given stage.

4.5. Pseudocode and mathematical example of Gain Ratioed Averaging (GRA)

Algorithm 1 Standard Gain Ratioed Averaging (GRA)

Require: Model predictions $\{P_1, P_2, \ldots, P_M\}$ ($P_i \in \mathbb{R}^{N\times K}$), true labels $Y \in \mathbb{N}^N$
Ensure: Ensemble predictions $\hat{Y}_{\text{ensemble}}$
Convert Y to one-hot encoding: $Y_{\text{oh}} \leftarrow \text{eye}(K)[Y]$
for each model $m \in 1, \ldots, M$ **do**
  Compute information gain:
  $\text{IG}_m \leftarrow H(Y) - H(Y \mid P_m)$
  Compute split information:
  $\text{SI}_m \leftarrow -\sum_{j=1}^{K} p_{m,j} \log p_{m,j}$
  Compute gain ratio:
  $\text{GR}_m \leftarrow \frac{\text{IG}_m}{\text{SI}_m + \epsilon}$
end for
Normalize weights:
$w_m \leftarrow \frac{\text{GR}_m}{\sum_{i=1}^{M} \text{GR}_i}$
$P_{\text{ensemble}} \leftarrow \sum_{m=1}^{M} w_m P_m$
return $\hat{Y}_{\text{ensemble}} \leftarrow \text{argmax}(P_{\text{ensemble}})$

4.5.1. Mathematical example

For a 3-class classification problem with the following model predictions:

$$P_1 = \begin{bmatrix} 0.9 & 0.1 & 0.0 \\ 0.0 & 0.8 & 0.2 \\ 0.7 & 0.3 & 0.0 \end{bmatrix}, \quad P_2 = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.8 & 0.1 & 0.1 \end{bmatrix}$$

True labels are given by:

$$Y = [0, 1, 0], \quad Y_{\text{oh}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

The entropy of the true labels is:

$$H(Y) = -\sum_{c=1}^{3} p(c)\log p(c)$$

For Model 1, the conditional entropy $H(Y \mid P_1)$ is low due to confident and correct predictions, resulting in a higher information gain:

$$\text{IG}_1 = H(Y) - H(Y \mid P_1)$$

Similarly, the split information is computed as:

$$\text{SI}_1 = -\sum_{j=1}^{3} \bar{p}_{1,j} \log \bar{p}_{1,j}$$

Thus, the gain ratio for Model 1 is:

$$\text{GR}_1 = \frac{\text{IG}_1}{\text{SI}_1}$$

Repeating the same steps for Model 2 yields GR_2, where $\text{GR}_1 > \text{GR}_2$ due to superior class separability.

Final normalized ensemble weights are computed as:

$$w_1 = \frac{\text{GR}_1}{\text{GR}_1 + \text{GR}_2}, \quad w_2 = \frac{\text{GR}_2}{\text{GR}_1 + \text{GR}_2}$$

The final ensemble prediction is obtained via a weighted average of softmax outputs followed by an argmax operation.



Fig. 7. Illustration of the Gain Ratioed Averaging (GRA) method in Stage-*p*.

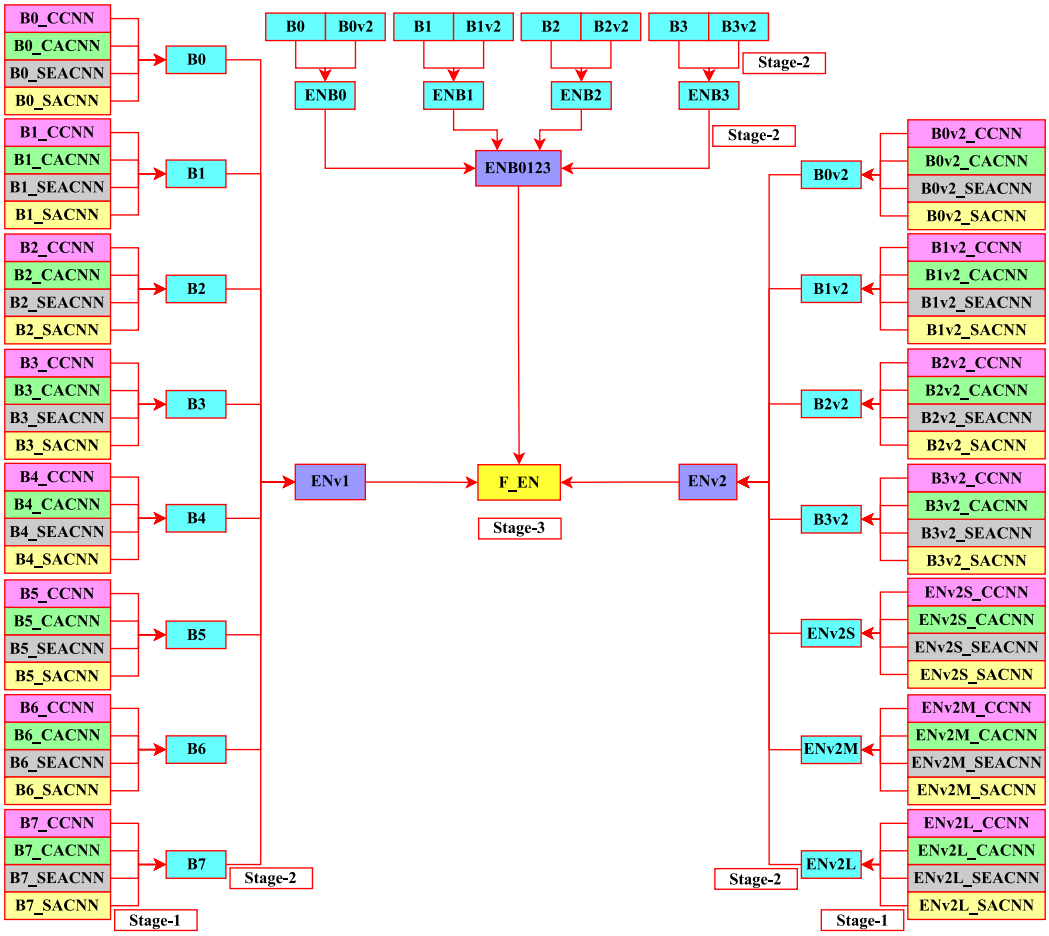


Fig. 8. Overall structure of the proposed Triple-Stage Gain Ratioed Averaging (TS-GRA).

4.6. Triple-stage GRA (TS-GRA)

To further strengthen the proposed GRA framework, we extend it into a *three-stage ensemble strategy*, referred to as **TS-GRA**. The primary motivation is to effectively highlight consistently superior models, even when their individual weights appear relatively small in earlier ensemble rounds. By structuring the aggregation sequentially across stages, models that demonstrate strong performance at earlier levels are progressively amplified in subsequent stages. Fig. 8 illustrates the complete organization of TS-GRA.

4.6.1. Stage 1: Base-level GRA

In the first stage, predictions from the four base architectures — CCNN, CACNN, SEACNN, and SACNN — are ensembled separately using GRA. Each classifier is trained on multiple EfficientNet backbones, resulting in a total of fifteen outputs. These include EfficientNet-v1 models (b0–b7) and EfficientNet-v2 variants (b0v2, b1v2, b2v2, b3v2, ENv2S, ENv2M, and ENv2L).

4.6.2. Stage 2: Intermediate GRA

The fifteen predictions obtained in Stage 1 are consolidated into three meta-level predictions by grouping EfficientNet variants:

- **ENv1:** Aggregates predictions from b0 through b7 (EfficientNet-v1).
- **ENv2:** Aggregates predictions from b0v2 through ENv2L (EfficientNet-v2).
- **ENb0123:** Combines matched pairs across both versions, i.e., (b0 + b0v2), (b1 + b1v2), (b2 + b2v2), and (b3 + b3v2).

This stage ensures that both intra-version consistency and cross-version complementarities are exploited.

4.6.3. Stage 3: Final GRA

In the last stage, the three aggregated predictions from Stage 2 (ENv1, ENv2, and ENb0123) are ensembled once more through GRA

to yield the **final ensemble output**, denoted as F_{EN}. This hierarchical design guarantees that high-performing models retain their influence across all stages, while weaker predictions are gradually down-weighted, thereby maximizing overall predictive accuracy and robustness.

4.7. Novelty over existing Tri-Attention models and hybrid EfficientNet ensembles

To the best of our knowledge, existing studies have not explored the integration of three complementary attention mechanisms — channel-wise attention, squeeze-and-excitation, and soft attention — within a single unified architecture. The proposed tri-attention design enables simultaneous modeling of inter-channel dependencies, spatial relevance, and adaptive feature recalibration, leading to richer and more discriminative representations.

Furthermore, prior hybrid EfficientNet ensemble approaches typically employ uniform input resolutions across all constituent models, limiting scalability and architectural diversity. In contrast, our framework leverages EfficientNet variants with heterogeneous input image sizes, enhancing model diversity and improving ensemble robustness.

Most importantly, the proposed TS-GRA introduces a gain-ratio–driven weighting strategy that adaptively assigns ensemble weights based on each model's discriminative contribution. Unlike traditional ensemble methods that rely on static or heuristic weighting schemes, TS-GRA dynamically emphasizes more informative models, resulting in consistently improved and stable performance across ensemble stages.

4.8. Performance evaluation metrics

The performance of the proposed models is evaluated using a diverse set of classification metrics, including accuracy, precision, recall (sensitivity), F1-score, specificity, and ROC-AUC (Receiver Operating Characteristic–Area Under the Curve) [34]. These measures collectively offer a balanced and detailed view of model effectiveness [35]. Each metric is computed based on values from the confusion matrix, which organizes model predictions into true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) [36].

$$\text{Accuracy (Acc)} = \frac{TP+TN}{TP+TN+FP+FN} \tag{13}$$

$$\text{Precision (Pre)} = \frac{TP}{TP+FP} \tag{14}$$

$$\text{Recall (Re)} = \frac{TP}{TP+FN} \tag{15}$$

$$\text{F1-Score (F1)} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{16}$$

$$\text{Specificity (Spe)} = \frac{TN}{TN+FP} \tag{17}$$

$$\text{True Positive Rate (TPR)} = \frac{TP}{TP+FN} \tag{18}$$

$$\text{False Positive Rate (FPR)} = \frac{FP}{FP+TN} \tag{19}$$

The ROC curve is plotted using TPR against FPR, and its area (ROC-AUC) provides a threshold-independent evaluation of model discriminative power. By analyzing these metrics collectively, we obtain a detailed understanding of model performance, which is essential for validating their suitability in real-world skin lesion classification scenarios.

4.9. Experimental setup

The proposed framework was developed and executed within Kaggle notebooks, utilizing a P100 GPU in combination with a 2-core Intel Xeon CPU, achieving an average computation rate of 690 ms per step. The input data comprised dermoscopic lesion images, each resized to (image size, image size, 3), with image_size differing between models, as outlined in Table 2. The dataset was partitioned into training (70%), validation (15%), and testing (15%) subsets using the `split-folders` library with its default random seed value of 1337 to ensure reproducibility. Model training was conducted over 50 epochs with a batch size of 16, using the Adam optimizer initialized with a learning rate of 0.001. Categorical cross-entropy was adopted as the objective loss, while convergence was regulated through early stopping supported by the ReduceLROnPlateau scheduler, applied with a patience threshold of 25 epochs. For reproducibility, the dataset has been made publicly available on Kaggle [37].

Table 3
Number of trainable parameters across all EfficientNet-based architectures.

Algorithms	Number of trainable parameters			
	CCNN	CACNN	SEACNN	SACNN
ENb0	13,240,067	13,282,659	20,127,109	19,095,331
ENb1	15,745,703	15,788,295	22,632,745	21,600,967
ENb2	13,894,409	13,937,001	20,781,451	16,603,945
ENb3	19,935,023	19,977,615	26,822,065	25,790,287
ENb4	26,793,679	26,836,271	33,680,721	32,648,943
ENb5	37,592,119	37,634,711	44,479,161	43,447,383
ENb6	49,993,311	50,035,903	56,880,353	55,848,575
ENb7	73,050,839	73,093,431	79,937,881	78,906,103
ENb0v2	15,091,223	15,133,815	21,978,265	20,946,487
ENb1v2	16,092,571	16,135,163	22,979,613	21,947,835
ENb2v2	14,880,501	14,923,093	21,767,543	17,590,037
ENb3v2	22,060,197	22,102,789	28,947,239	27,915,461
ENv2S	29,410,007	29,452,599	36,297,049	35,265,271
ENv2M	62,090,875	62,133,467	68,977,917	67,946,139
ENv2L	126,466,791	126,509,383	133,353,833	132,322,055

Table 4
Training Time (s/epoch) across all EfficientNet-based architectures.

Algorithms	Training time (Seconds/Epoch)			
	CCNN	CACNN	SEACNN	SACNN
ENb0	432	440	492	501
ENb1	440	456	510	519
ENb2	449	460	521	530
ENb3	458	467	530	560
ENb4	466	472	538	582
ENb5	480	486	549	592
ENb6	488	498	568	601
ENb7	497	510	577	609
ENb0v2	412	415	436	439
ENb1v2	430	435	441	449
ENb2v2	482	488	499	508
ENb3v2	592	593	601	619
ENv2S	430	433	449	456
ENv2M	588	591	602	620
ENv2L	629	635	648	661

4.9.1. Trainable parameters and training time

Since the GRA ensemble is applied at the prediction stage, the number of trainable parameters remains unchanged across individual models. Table 3 details the parameter counts for each architecture.

Notably, ENv2L contains the largest number of parameters, reaching approximately 133 million under the SEACNN configuration. In contrast, other variants contain considerably fewer parameters. Since each model operates independently before ensemble integration, GRA preserves efficient prediction times without introducing additional computational overhead.

Additionally, Table 4 summarizes the time required to train each model. Because all models are independent of one another, they can be trained concurrently; therefore, parallel execution does not introduce any additional overall training time. Moreover, since all experiments were conducted using Kaggle notebooks, no extra memory resources were required.

From the results, ENv2L (SACNN) exhibits the highest training cost, requiring 661 s per epoch. When all models are trained in parallel,

the total training time is effectively bounded by this model, while the remaining models finish earlier. Consequently, from a computational perspective, the proposed setup incurs no additional overhead.

4.9.2. Hyperparameter selection

Hyperparameters were determined through manual tuning, guided by empirical evaluation and established deep learning practices. Key considerations included the learning rate, batch size, and architectural components such as kernel initializers and activation functions. The Adam optimizer was configured with a learning rate of 0.001 to enable stable yet adaptive weight updates. Batch normalization was employed to stabilize activations, improving convergence and mitigating overfitting. We applied the `He normal` initializer to facilitate efficient gradient propagation during training. ReLU activations were chosen for their ability to capture complex nonlinear relationships, thereby improving classification accuracy. This careful tuning ensured a balance between computational efficiency and state-of-the-art performance in skin lesion classification.

5. Experimental results analysis

This section provides a detailed examination of the theoretical foundations together with graphical illustrations to assess classification outcomes. The central objective is to demonstrate how the integration of GRA enhances the predictive accuracy of the SEN framework. To support this claim, experimental evidence is reported through a variety of evaluation measures and visual tools — including ROC-AUC plots and confusion matrices — which enable a clear comparison among the different strategies introduced earlier.

In this context, the application of GRA across all classifiers at each stage, denoted as GRA_i (with i indicating the stage), is employed to construct the $TS\text{-}GRA$.

5.1. SEN architectures in Stage 1

A total of fifteen models, as previously mentioned, are employed, each paired with CCNN, CACNN, SEACNN, and SACNN for every model variant. The outcomes derived from these various combinations, along with the results from Stage-1 GRA, are presented in Tables 5 to 8.

Tables 5 and 6 presents a detailed analysis of the classification performance of various attention mechanisms integrated into all EfficientNet version-1 architectures (B0 through B7). The primary attention mechanisms evaluated are CCNN, CACNN, SEACNN, SACNN, and GRA_1 is the ensembled performances of them.

In the B0 architecture, GRA achieves the highest accuracy of 93.24%, significantly outperforming CCNN (92.03%), CACNN (91.30%), SEACNN (91.79%), and SACNN (91.18%). Additionally, GRA maintains the highest precision (92.74%) and recall (93.24%), resulting in the best F1-score (92.78%). The specificity of 84.32% further indicates GRA's robust ability to correctly identify negative cases.

For the B1 architecture, GRA again leads with an accuracy of 93.12%, surpassing CCNN (92.63%), CACNN (91.67%), SEACNN (91.30%), and SACNN (90.82%). The precision (92.84%), recall (93.12%), and F1-score (92.68%) metrics further emphasize GRA's consistent excellence. The specificity for GRA is 83.82%, maintaining its strong performance across metrics.

In B2, GRA achieves an accuracy of 93.12%, continuing its dominance over CCNN (92.51%), CACNN (91.43%), SEACNN (92.27%), and SACNN (91.30%). The precision of 92.65%, recall of 93.12%, and F1-score of 92.65% highlight GRA's superiority. The specificity stands at 85.25%, demonstrating balanced performance across all metrics.

The B3 architecture sees GRA attaining the highest accuracy of 93.36%, outshining CCNN (92.63%), CACNN (92.15%), SEACNN (91.55%), and SACNN (92.03%). With a precision of 93.01%, recall of 93.36%, and F1-score of 93.00%, GRA's metrics underscore its top-tier performance. The specificity for GRA is 85.29%, reflecting its reliable classification capabilities.

Table 5
Performance Metrics of Stage-1 GRA (EfficientNetB0 - EfficientNetB3).

Algorithm	Accuracy	Precision	Recall	F1-Score	Specificity
B0_CCNN	92.03	91.26	92.03	91.45	81.87
B0_CACNN	91.30	90.66	91.30	90.85	82.77
B0_SEACNN	91.79	91.24	91.79	91.20	80.88
B0_SACNN	91.18	91.18	91.18	91.02	86.12
GRA_1 (B0)	93.24	92.74	93.24	92.78	84.32
B1_CCNN	92.63	92.45	92.63	92.35	85.69
B1_CACNN	91.67	91.30	91.67	91.23	82.77
B1_SEACNN	91.30	90.39	91.30	90.48	77.98
B1_SACNN	90.82	90.70	90.82	90.69	86.61
GRA_1 (B1)	93.12	92.84	93.12	92.68	83.82
B2_CCNN	92.51	91.95	92.51	92.05	84.28
B2_CACNN	91.43	90.58	91.43	90.75	82.28
B2_SEACNN	92.27	91.92	92.27	91.76	84.72
B2_SACNN	91.30	90.98	91.30	91.10	86.10
GRA_1 (B2)	93.12	92.65	93.12	92.65	85.25
B3_CCNN	92.63	92.51	92.63	92.41	86.70
B3_CACNN	92.15	91.61	92.15	91.65	81.87
B3_SEACNN	91.55	91.07	91.55	91.05	80.90
B3_SACNN	92.03	91.88	92.03	91.78	84.72
GRA_1 (B3)	93.36	93.01	93.36	93.00	85.29

In B4, GRA maintains an accuracy of 92.87%, leading over CCNN (92.15%), CACNN (92.39%), SEACNN (91.43%), and SACNN (92.03%). The precision (92.46%), recall (92.87%), and F1-score (92.45%) of GRA continue to be the highest among all tested mechanisms. GRA's specificity is 84.74%, ensuring its effectiveness in identifying true negatives.

For B5, GRA achieves an accuracy of 92.99%, ahead of CCNN (92.15%), CACNN (92.27%), SEACNN (91.91%), and SACNN (91.55%). The precision of 92.51%, recall of 92.99%, and F1-score of 92.55% illustrate GRA's consistent superiority. The specificity for GRA is 85.71%, further confirming its balanced performance.

In B6, GRA holds the highest accuracy of 93.12%, outperforming CCNN (91.79%), CACNN (92.27%), SEACNN (92.87%), and SACNN (91.06%). With a precision of 92.82%, recall of 93.12%, and F1-score of 92.73%, GRA continues to lead across metrics. The specificity for GRA is 84.78%, indicating its robustness in classification tasks.

Even in the competitive B7 architecture, GRA maintains an accuracy of 92.75%, surpassing CCNN (92.51%), CACNN (91.06%), SEACNN (90.82%), and SACNN (91.91%). The precision (92.18%), recall (92.75%), and F1-score (92.31%) of GRA are the highest among all mechanisms. The specificity stands at 84.76%, underscoring GRA's balanced and reliable performance.

Overall, GRA demonstrates its superiority across all EfficientNet architectures by consistently achieving the highest score of all metrics. Its robust performance highlights its effectiveness in classification tasks, making it a valuable addition to the EfficientNet architectures.

Tables 7 and 8 provides a comprehensive evaluation of the classification performance of various attention mechanisms integrated into EfficientNetV2 architectures (B0v2, B1v2, B2v2, B3v2, ENv2S, ENv2M, and ENv2L). The attention mechanisms evaluated are CCNN, CACNN, SEACNN, SACNN, and GRA_1 is the averaged performances of them.

In the B0v2 architecture, GRA achieves the highest Acc of 93.12%, outperforming CCNN (91.06%), CACNN (90.34%), SEACNN (92.15%), and SACNN (91.18%). GRA also excels in precision (92.71%), recall (93.12%), and F1-score (92.63%), demonstrating its overall superior performance. The specificity for GRA is 84.32%, indicating robust performance in identifying negative cases.

For the B1v2 architecture, GRA again leads with an accuracy of 93.96%, significantly higher than CCNN (92.15%), CACNN (92.15%), SEACNN (92.03%), and SACNN (92.39%). GRA's precision (93.66%),

Table 6
Performance Metrics of Stage-1 GRA (EfficientNetB4 - EfficientNetB7).

Algorithm	Accuracy	Precision	Recall	F1-Score	Specificity
B4_CCNN	92.15	91.76	92.15	91.73	83.29
B4_CACNN	92.39	92.47	92.39	92.35	91.90
B4_SEACNN	91.43	91.19	91.43	91.14	85.14
B4_SACNN	92.03	91.51	92.03	91.51	80.92
GRA_1 (B4)	92.87	92.46	92.87	92.45	84.74
B5_CCNN	92.15	92.41	92.15	92.16	90.47
B5_CACNN	92.27	91.70	92.27	92.00	86.64
B5_SEACNN	91.91	91.44	91.91	91.53	84.24
B5_SACNN	91.55	90.88	91.55	91.08	83.25
GRA_1 (B5)	92.99	92.51	92.99	92.55	85.71
B6_CCNN	91.79	91.40	91.79	91.44	84.69
B6_CACNN	92.27	91.92	92.27	91.95	83.31
B6_SEACNN	92.87	92.56	92.87	92.43	84.77
B6_SACNN	91.06	90.88	91.06	90.83	88.45
GRA_1 (B6)	93.12	92.82	93.12	92.73	84.78
B7_CCNN	92.51	91.93	92.51	92.04	86.65
B7_CACNN	91.06	90.05	91.06	90.31	80.36
B7_SEACNN	90.82	90.85	90.82	90.82	87.04
B7_SACNN	91.91	91.33	91.91	91.30	81.38
GRA_1 (B7)	92.75	92.18	92.75	92.31	84.76

Table 7
Performance Metrics of Stage-1 GRA (EfficientNetB0v2 - EfficientNetB3v2.

Algorithm	Accuracy	Precision	Recall	F1-Score	Specificity
B0v2_CCNN	91.06	90.40	91.06	90.58	82.77
B0v2_CACNN	90.34	89.44	90.34	89.33	77.49
B0v2_SEACNN	92.15	91.96	92.15	91.94	87.61
B0v2_SACNN	91.18	90.70	91.18	90.64	82.30
GRA_1 (B0v2)	93.12	92.71	93.12	92.63	84.32
B1v2_CCNN	92.15	91.63	92.15	91.77	84.25
B1v2_CACNN	92.15	91.66	92.15	91.38	81.36
B1v2_SEACNN	92.03	91.63	92.03	91.65	86.67
B1v2_SACNN	92.39	91.86	92.39	91.89	82.85
GRA_1 (B1v2)	93.96	93.66	93.96	93.53	86.27
B2v2_CCNN	91.55	91.03	91.55	91.04	81.37
B2v2_CACNN	92.39	92.01	92.39	91.86	81.43
B2v2_SEACNN	91.67	91.15	91.67	91.16	80.39
B2v2_SACNN	91.91	91.41	91.91	91.55	86.18
GRA_1 (B2v2)	93.24	92.81	93.24	92.74	81.93
B3v2_CCNN	91.67	91.29	91.67	91.39	86.17
B3v2_CACNN	92.51	91.84	92.51	92.00	82.84
B3v2_SEACNN	92.27	91.69	92.27	91.76	81.40
B3v2_SACNN	91.79	91.68	91.79	91.71	88.52
GRA_1 (B3v2)	93.48	93.01	93.48	93.10	85.29

Table 8
Performance Metrics of Stage-1 GRA (EfficientNetv2S - EfficientNetB3v2L.

Algorithm	Accuracy	Precision	Recall	F1-Score	Specificity
ENv2S_CCNN	91.55	91.02	91.55	91.05	80.39
ENv2S_CACNN	92.39	92.02	92.39	92.03	86.63
ENv2S_SEACNN	91.18	90.88	91.18	90.65	81.83
ENv2S_SACNN	93.12	92.71	93.12	92.74	86.70
GRA_1 (ENv2S)	93.60	93.22	93.60	93.24	85.77
ENv2M_CCNN	92.63	92.23	92.63	92.14	84.72
ENv2M_CACNN	91.67	91.49	91.67	91.34	83.31
ENv2M_SEACNN	91.30	91.01	91.30	90.77	82.21
ENv2M_SACNN	91.91	91.48	91.91	91.56	85.69
GRA_1 (ENv2M)	92.75	92.21	92.75	92.25	83.78
ENv2L_CCNN	92.51	92.19	92.51	92.22	88.60
ENv2L_CACNN	93.72	93.53	93.72	93.52	88.18
ENv2L_SEACNN	93.12	92.77	93.12	93.13	87.16
ENv2L_SACNN	93.84	93.61	93.84	93.54	90.07
GRA_1 (ENv2L)	94.20	93.96	94.20	93.82	85.80

recall (93.96%), and F1-score (93.53%) underscore its consistent excellence. The specificity for GRA is 86.27%, maintaining strong performance across metrics.

In B2v2, GRA achieves an accuracy of 93.24%, continuing its dominance over CCNN (91.55%), CACNN (92.39%), SEACNN (91.67%), and SACNN (91.91%). The precision of 92.81%, recall of 93.24%, and F1-score of 92.74% highlight GRA's superior classification capabilities. The specificity stands at 81.93%, ensuring balanced performance across all metrics.

The B3v2 architecture sees GRA attaining the highest accuracy of 93.48%, surpassing CCNN (91.67%), CACNN (92.51%), SEACNN (92.27%), and SACNN (91.79%). With a precision of 93.01%, recall of 93.48%, and F1-score of 93.10%, GRA's metrics underscore its top-tier performance. The specificity for GRA is 85.29%, reflecting its reliable classification capabilities.

In the ENv2S architecture, GRA maintains an accuracy of 93.60%, leading over CCNN (91.55%), CACNN (92.39%), SEACNN (91.18%), and SACNN (93.12%). The precision (93.22%), recall (93.60%), and F1-score (93.24%) of GRA continue to be the highest among all tested mechanisms. GRA's specificity is 85.77%, ensuring its effectiveness in identifying true negatives.

For ENv2M, GRA achieves an accuracy of 92.75%, ahead of CCNN (92.63%), CACNN (91.67%), SEACNN (91.30%), and SACNN (91.91%). The precision of 92.21%, recall of 92.75%, and F1-score of 92.25% illustrate GRA's consistent superiority. The specificity for GRA is 83.78%, further confirming its balanced performance.

In the ENv2L architecture, GRA holds the highest accuracy of 94.20%, outperforming CCNN (92.51%), CACNN (93.72%), SEACNN (93.12%), and SACNN (93.84%). With a precision of 93.96%, recall of 94.20%, and F1-score of 93.82%, GRA continues to lead across metrics. The specificity for GRA is 85.80%, indicating its robustness in classification tasks.

Overall, GRA demonstrates its superiority across all EfficientNetV2 architectures by consistently achieving the highest accuracy, precision, recall, and F1-scores. Its robust performance highlights its effectiveness in classification tasks, making it a valuable addition to the EfficientNetV2 architectures.

5.2. SEN architectures in Stage 2

At Stage 2, our methodology leverages distinct combinations derived from the Stage-1 predictions. Specifically, we perform two separate fusions: one for the EfficientNetV1 (ENv1) models and another for the EfficientNetV2 (ENv2) models. The ENv1 combination integrates predictions from GRA-augmented EfficientNetV1 models (B0 to B7), whereas the ENv2 combination consolidates the results from GRA-enhanced EfficientNetV2 models (B0v2, B1v2, B2v2, B3v2, ENv2S, ENv2M, ENv2L). Finally, the composite 'ENb0123' amalgamation consolidates the predictive abilities of various EfficientNet architectures by combining their common variants. Specifically, the ensemble of EfficientNetV1 B0 and EfficientNetV2 B0v2 generates ENb0; B1 and B1v2 create ENb1; B2 and B2v2 result in ENb2; and B3 combined with B3v2 makes the ensembled ENb3 predictions. These four predictions are further combined to obtain the ENb0123 prediction. The results of these combinations are outlined in Table 9.

The fusion of ENv1 models demonstrates an overall improvement in performance metrics. The Acc achieved by the combined ENv1 models is 93.48%, which is higher than any individual ENv1 model, indicating the effectiveness of the fusion approach. This fusion also attains a precision of 93.08%, recall of 93.48%, F1-score of 93.07%, and specificity of 85.27%. The GRA-enhanced B3 model within the ENv1 set shows the highest individual accuracy of 93.36%, followed closely by other models, reinforcing the robustness of the ensemble method.

The ENv2 models display superior performance metrics when combined. The ENv2 fusion achieves an impressive accuracy of 94.44%, significantly higher than the individual models, demonstrating the

Table 9
Performance Metrics of Stage - 2 GRA (ENv1, ENv2, ENB0123)

Algorithm	Accuracy	Precision	Recall	F1-Score	Specificity
GRA_1 (B0)	93.24	92.74	93.24	92.78	84.32
GRA_1 (B1)	93.12	92.84	93.12	92.68	83.82
GRA_1 (B2)	93.12	92.65	93.12	92.65	85.25
GRA_1 (B3)	93.36	93.01	93.36	93.00	85.29
GRA_1 (B4)	92.87	92.46	92.87	92.45	84.74
GRA_1 (B5)	92.99	92.51	92.99	92.55	85.71
GRA_1 (B6)	93.12	92.82	93.12	92.73	84.78
GRA_1 (B7)	92.75	92.18	92.75	92.31	84.76
GRA_2 (ENv1)	93.48	93.08	93.48	93.07	85.27
GRA_1 (B0v2)	93.12	92.71	93.12	92.63	84.32
GRA_1 (B1v2)	93.96	93.66	93.96	93.53	86.27
GRA_1 (B2v2)	93.24	92.81	93.24	92.74	81.93
GRA_1 (B3v2)	93.48	93.01	93.48	93.10	85.29
GRA_1 (ENv2S)	93.60	93.22	93.60	93.24	85.77
GRA_1 (ENv2M)	92.75	92.21	92.75	92.25	83.78
GRA_1 (ENv2L)	94.44	94.17	94.44	94.21	90.13
GRA_2 (ENv2)	94.44	94.17	94.44	94.21	90.13
GRA_1 (B0)	93.24	92.74	93.24	92.78	84.32
GRA_1 (B0v2)	93.12	92.71	93.12	92.63	84.32
GRA_2 (ENB0)	93.48	93.18	93.48	93.04	84.34
GRA_1 (B1)	93.12	92.84	93.12	92.68	83.82
GRA_1 (B1v2)	93.96	93.66	93.96	93.53	86.27
GRA_2 (ENB1)	93.84	93.62	93.84	93.33	84.35
GRA_1 (B2)	93.12	92.65	93.12	92.65	85.25
GRA_1 (B2v2)	93.24	92.81	93.24	92.74	81.93
GRA_2 (ENB2)	93.84	93.55	93.84	93.45	84.82
GRA_1 (B3)	93.36	93.01	93.36	93.00	85.29
GRA_1 (B3v2)	93.48	93.01	93.48	93.10	85.29
GRA_2 (ENB3)	94.08	93.85	94.08	93.75	86.30
GRA_2 (ENB0123)	94.20	93.92	94.20	93.80	84.83

strength of the fusion approach. The precision, recall, and F1-score for the ENv2 fusion are 94.17%, 94.44%, and 94.21%, respectively, with a specificity of 90.13%. Notably, the ENv2L model individually achieves the promising accuracy of 94.20%, showcasing the powerful performance of GRA when applied to larger models.

The fusion of GRA-enhanced models at Stage 2, both for EfficientNetV1 and EfficientNetV2, results in improved performance metrics across the board. The ENv2 fusion outperforms the ENv1 fusion, highlighting the advancements in the EfficientNetV2 architecture. The results underscore the superiority of the GRA mechanism, which consistently enhances the classification performance of both EfficientNet versions. These findings affirm the efficacy of our methodology in utilizing model combinations to achieve higher accuracy, precision, recall, F1-scores, and specificity, thereby proving the robustness of the proposed approach.

The GRA-based B0 variant secures an accuracy of 93.24%, while its updated version, B0v2, attains 93.12%. When these models are integrated, the resulting ENb0 ensemble slightly raises performance, achieving 93.48% accuracy together with 93.18% precision, 93.48% recall, an F1-score of 93.04%, and a specificity of 84.34%.

For the B1 family, the baseline records 93.12% accuracy, whereas B1v2 improves to 93.96%. The ensemble ENb1 maintains this upward trend, reaching 93.84% accuracy, along with 93.62% precision, 93.84% recall, an F1-score of 93.33%, and a specificity of 84.35%.

The B2 and B2v2 models also perform consistently, achieving accuracies of 93.12% and 93.24%, respectively. Their ensemble, ENb2, surpasses both with 93.84% accuracy, 93.55% precision, 93.84% recall, an F1-score of 93.45%, and 84.82% specificity.

Finally, the GRA-enhanced B3 version yields 93.36% accuracy, while B3v2 slightly edges ahead at 93.48%. The ensemble ENb3 delivers the best performance among them, achieving 94.08% accuracy, 93.85% precision, 94.08% recall, an F1-score of 93.75%, and a specificity of 86.30%.

Table 10
Performance Metrics of Stage - 3 GRA (F_EN).

Algorithm	Accuracy	Precision	Recall	F1-Score	Specificity
GRA_2					
ENv1	93.48	93.08	93.48	93.07	85.27
ENv2	94.44	94.17	94.44	94.21	90.13
ENb0123	94.20	93.92	94.20	93.80	84.83
GRA_3 (F_EN)	94.57	94.18	94.57	94.23	88.69

The final composite ENb0123, which integrates the predictions from ENb0, ENb1, ENb2, and ENb3, achieves an overall accuracy of 94.20%. The precision, recall, and F1-score are 93.92%, 94.20%, and 93.80%, respectively, with a specificity of 84.83%. This result highlights the strength of combining various EN architectures to enhance predictive performance.

The amalgamation of predictions from various EfficientNet architectures, especially through the ENb0123 composite model, demonstrates the effectiveness of ensemble methods in improving classification performance. The combined models consistently outperform their individual counterparts, showcasing the robustness and enhanced accuracy achieved through the GRA mechanism and model fusion approach. The results affirm the superior performance of the proposed methodology, underscoring its potential for practical applications in various classification tasks.

5.3. SEN architectures in Stage 3

The performance metrics of the Final EfficientNet (F_EN) ensemble, as displayed in Table 10, underscore its superiority in predictive accuracy and reliability. The F_EN ensemble achieves an impressive accuracy of 94.57%, marking a significant enhancement over individual models from previous stages. This high level of accuracy reflects the robustness of the ensemble method, ensuring that the predictions are consistently close to the actual outcomes.

In terms of precision, the F_EN ensemble records a score of 94.18%, indicating its effectiveness in minimizing false positives. This high precision rate suggests that the ensemble is particularly adept at identifying relevant instances accurately, thereby reducing the occurrence of incorrect classifications.

The recall metric for the F_EN ensemble stands at 94.57%, demonstrating its ability to capture a high proportion of actual positive cases. This high recall rate is crucial for applications where identifying all relevant cases is essential, showcasing the ensemble's capability to detect true positives reliably.

The F1-score, which balances precision and recall, further validates the F_EN ensemble's performance with a score of 94.23%. This balanced performance metric indicates that the ensemble not only excels in accuracy but also maintains a harmonious trade-off between precision and recall, ensuring comprehensive and reliable predictions.

Lastly, the specificity of the F_EN ensemble is recorded at 88.69%, highlighting its competence in correctly identifying negative cases. This high specificity underscores the ensemble's efficiency in avoiding false negatives, thus enhancing its reliability in diverse classification tasks.

In summary, the superior performance metrics of the F_EN by $TS-GRA$ demonstrate its exceptional capability in predictive modeling, positioning it as a highly effective and reliable tool in the realm of skin lesion classification.

Table 11
Performance Metrics of Pre-final and Final Stages with 95% Confidence Intervals.

Algorithm	Accuracy	Precision	Recall	F1-Score	Specificity
GRA_2 (ENv1)	93.48 ± 1.47	93.08 ± 1.51	93.48 ± 1.47	93.07 ± 1.51	85.27 ± 2.10
GRA_2 (ENv2)	94.44 ± 1.36	94.17 ± 1.39	94.44 ± 1.36	94.21 ± 1.39	90.13 ± 1.77
GRA_2 (ENB0123)	94.20 ± 1.39	93.92 ± 1.42	94.20 ± 1.39	93.80 ± 1.43	84.83 ± 2.13
GRA_3 (F_EN)	**94.57 ± 1.34**	**94.18 ± 1.39**	**94.57 ± 1.34**	**94.23 ± 1.39**	**88.69 ± 1.88**

5.4. Results with confidence interval

The comparative performance of the pre-final and final stages of the proposed architecture is summarized in Table 11, where each metric is reported with its corresponding 95% confidence interval (CI) based on a test size.

The results demonstrate that the final stage fusion model, GRA_3 (F_EN), achieves the highest overall performance across all primary metrics, specifically reaching an accuracy of 94.57% ± 1.34% and an F1-Score of 94.23% ± 1.39%. The narrow margins of error indicate high stability and statistical reliability in the model's predictive capabilities across the 7-class dataset. While the GRA_2 (ENv2) ensemble shows competitive specificity (90.13% ± 1.77%), the GRA_3 model maintains a superior balance between precision and recall. This suggests that the stage-3 fusion successfully leverages the complementary features of the preceding EfficientNet-based architectures to effectively minimize classification errors.

5.5. Performance analysis by visualization

5.5.1. Confusion matrix

Given the large number of classifiers and their respective variations, the confusion matrices of individual models at different stages of the GRA framework are not reported. Instead, emphasis is placed on the confusion matrices generated through the TS-GRA method, specifically from the second ensemble stage through to the final stage. These results, displayed in Figs. 9 and 12, provide a clear summary of classification accuracy and misclassification patterns across categories, while also confirming the strong predictive capability of TS-GRA in handling all classes effectively.

Fig. 9 depicts the Stage 2 performance of the ENv1 architecture, where class-specific trends become evident. For instance, the AK class achieves 13 correct detections but suffers from 9 errors, indicating moderate reliability. The BCC class demonstrates notable strength, with 22 accurate predictions and only 5 misclassifications. Similarly, BKL achieves solid results, correctly identifying 49 samples while misclassifying 17. DF shows a high success rate with 5 correct and only 1 incorrect classification. MEL, however, remains the most challenging class, with 19 correct cases offset by 16 errors, highlighting its tendency to overlap with other categories. The NV class exhibits outstanding performance, producing 657 accurate predictions with merely 6 errors. Lastly, VASC stands out by achieving flawless recognition across all 9 samples. Collectively, these outcomes emphasize the predominance of correct classifications, while the relatively limited number of errors reinforces the robustness of the approach, albeit with some room for refinement in the more challenging classes.

At Stage 2, the ENv2 ensemble exhibits a notable improvement compared to Stage 1, primarily reflected in the reduced rate of misclassifications. The corresponding confusion matrix (Fig. 10) highlights these gains across several classes. For AK, 12 samples are classified correctly while 10 are misclassified, indicating moderate performance with room for refinement. The BCC class shows strong reliability, with 22 accurate predictions and only 5 errors. Similarly, BKL achieves 53 correct identifications against 13 errors, suggesting solid recognition with a relatively small error margin. DF is also handled effectively, with 5 correct cases and just a single misclassification, reflecting consistent performance despite the limited sample size. MEL remains the most difficult category, yielding 23 accurate predictions alongside 12

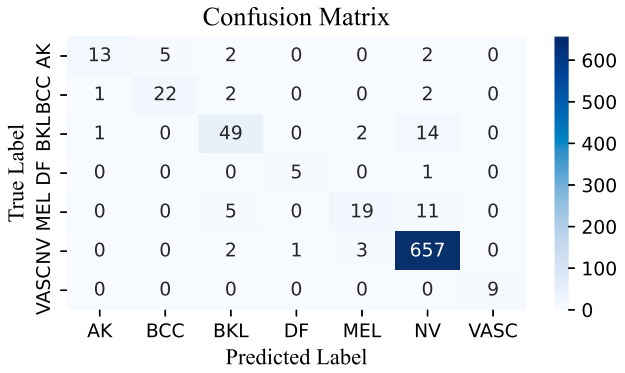


Fig. 9. Confusion matrix of ENv1 (TS-GRA Stage - 2)

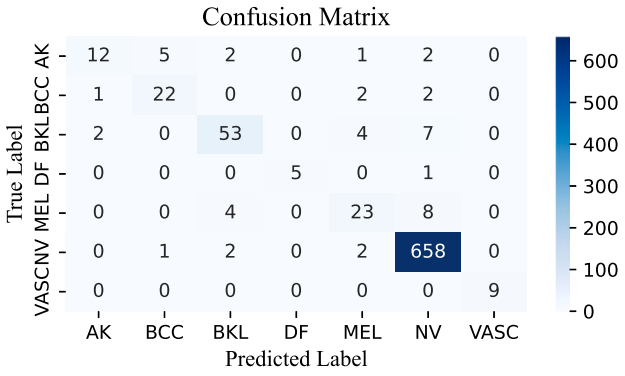


Fig. 10. Confusion matrix of ENv2 (TS-GRA Stage - 2)

misclassifications, which can be attributed to its overlap with features of other classes. In contrast, NV demonstrates excellent results, with 658 correct identifications and only 5 errors. Finally, VASC achieves flawless classification, with all 9 cases correctly recognized, reflecting the distinctiveness of its features. Overall, the dominance of true positives across categories emphasizes the robustness of ENv2 at Stage 2, while the relatively small number of errors further confirms its strong predictive ability and leaves scope for additional optimization.

Lastly, in Stage 2, the ENb0123 ensemble delivers strong results, though its overall performance leaves slightly more room for improvement when compared with the ENv2 model. The confusion matrix in Fig. 11 illustrates these outcomes in detail. For the AK class, 13 cases are classified correctly while 9 are misclassified, suggesting a need for better class separation. The BCC category achieves 23 accurate predictions with only 4 errors, highlighting robust recognition, albeit marginally weaker than ENv2. BKL attains 51 correct identifications with 15 misclassifications, demonstrating good capability but with some variability. DF is consistently detected with 5 true positives and just a single misclassification. MEL remains the most challenging class, producing 20 correct detections but accompanied by 15 errors, pointing to feature overlap with other categories. NV continues to dominate with 659 accurate predictions and only 4 errors, confirming the model's exceptional ability to identify this class. Finally, VASC achieves perfect recognition, with all 9 samples correctly classified, reflecting high precision due to distinct class characteristics. Taken together, the ENb0123

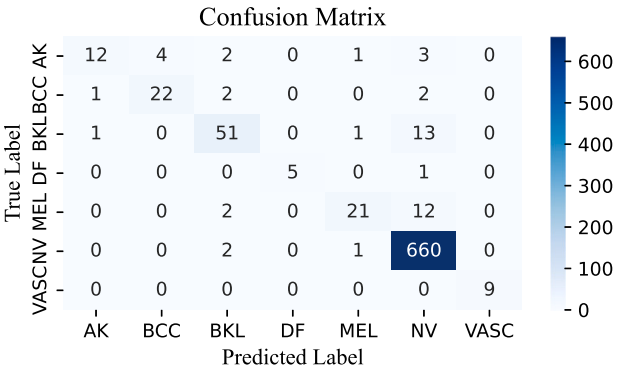


Fig. 11. Confusion matrix of ENb0123 (TS-GRA Stage - 2)

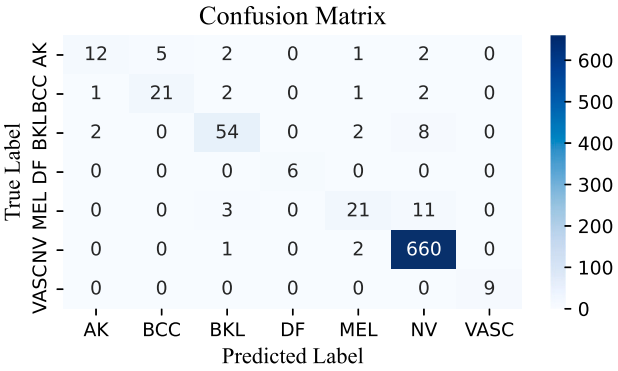


Fig. 12. Confusion Matrix of F_EN (TS-GRA Stage - 3)

ensemble shows reliable classification across categories, with strong true positive rates and clear opportunities for further refinement.

The final confusion matrix, presented in Fig. 12, offers a comprehensive view of classification performance across all categories. For the AK class, 12 samples are correctly classified, while 10 are misclassified into other categories, indicating that further improvement is still required. The BCC class achieves 21 true positives with only 6 errors, reflecting strong but not perfect recognition. BKL demonstrates 54 correct identifications against 12 misclassifications, underscoring the model's effectiveness in handling this category with a modest error margin. The DF class, although represented by a small number of cases, records 6 true positives with no misclassifications, showing reliable detection. MEL continues to present challenges, yielding 21 correct predictions alongside 14 errors, which may stem from overlapping features with other classes. NV emerges as the best-performing class, with 660 correct identifications and only 3 errors, highlighting the model's high proficiency in recognizing this category. VASC achieves flawless classification, with all 9 samples correctly identified and no misclassifications, likely benefiting from distinct feature characteristics and limited sample size.

Overall, the predominance of diagonal values in the confusion matrix points to strong predictive performance, while the relatively sparse off-diagonal entries indicate only minor misclassification issues. Nonetheless, the variation in class sample sizes suggests that precision and recall should be interpreted carefully, as they may be affected by imbalanced distributions.

5.5.2. Receiver Operating Characteristic Area Under Curve (ROC-AUC)

The ROC-AUC curve provides an effective means of visualizing and evaluating model performance. Accordingly, this metric is analyzed for the TS-GRA method. The graphical results, shown in Fig. 13, correspond to the final stage (Stage 3) of the framework. The curves exhibit very limited variation across the different classes, highlighting the stability of the model and its consistently strong predictive capability.

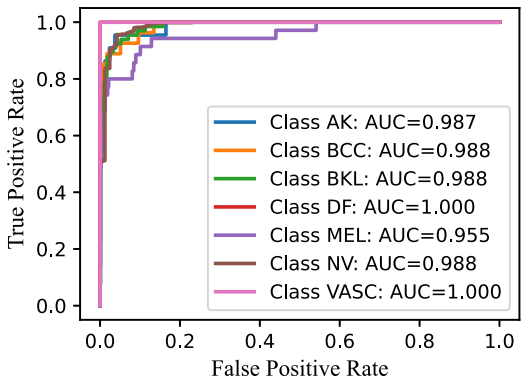


Fig. 13. ROC-AUC Curve obtained by TS-GRA (Stage - 3)

5.5.3. Gradient Class Activation Map (GradCAM)

The process begins with the generation of a heatmap from the original image, which forms the basis for the GradCAM visualization. This heatmap highlights the regions that the model considers most relevant [38]. From it, the GradCAM view is constructed, offering a clearer depiction of the image areas prioritized by the model. Fig. 14 presents these visualizations for all seven classes, where representative samples are displayed alongside their corresponding GradCAM outputs. The comparison between the original images and their GradCAM views demonstrates the model's ability to accurately localize the most informative regions, thereby contributing to improved classification performance.

Fig. 15 illustrates a representative image from the MEL class. The attention-driven architectures, particularly ENb0, successfully highlight the relevant region of interest, whereas SACNN does not perform as effectively for this example. The inclusion of additional models further strengthens our findings, as their integration consistently leads to correct predictions. This can be observed through the GradCAM visualizations with TA, which provide further evidence of the ensemble's reliability. A key advancement, however, arises from the GRA ensemble, which enables highly accurate class predictions. This outcome highlights the strength of the proposed $TS\text{-}GRA$ method, showcasing its ability to overcome the limitations of individual classifiers and establish superior predictive performance.

5.6. Ablation study

To highlight the advantages of the proposed framework over existing state-of-the-art approaches, a detailed ablation study is carried out. This analysis centers on two core innovations: Tri-Attention (TA) and Gain Ratioed Averaging (GRA). The contribution of each component is assessed by comparing model performance with and without their inclusion, thereby quantifying their impact on the overall results.

5.6.1. Utilization of GRA without TA

Our application of GRA across various EN models, including ENv1 and ENv2, in both TA-integrated and non-TA configurations, sounds thorough and methodical. The comparison presented in Table 12 provides valuable insights into the impact of attention mechanisms on performance. Highlighting the effectiveness of TA by contrasting results from configurations with and without attention modules is a smart approach. This clearly demonstrates the benefits of TA and supports the case for its inclusion in the proposed architecture. Attention to these details helps in making a compelling argument for the enhancements brought by TA.

The table clearly highlights the exceptional performance of our GRA_{F_EN} approach, which integrates attention modules (TA), compared to models that do not utilize TA. Our GRA_{F_EN} approach consistently outperforms all other configurations in terms of all metrics.

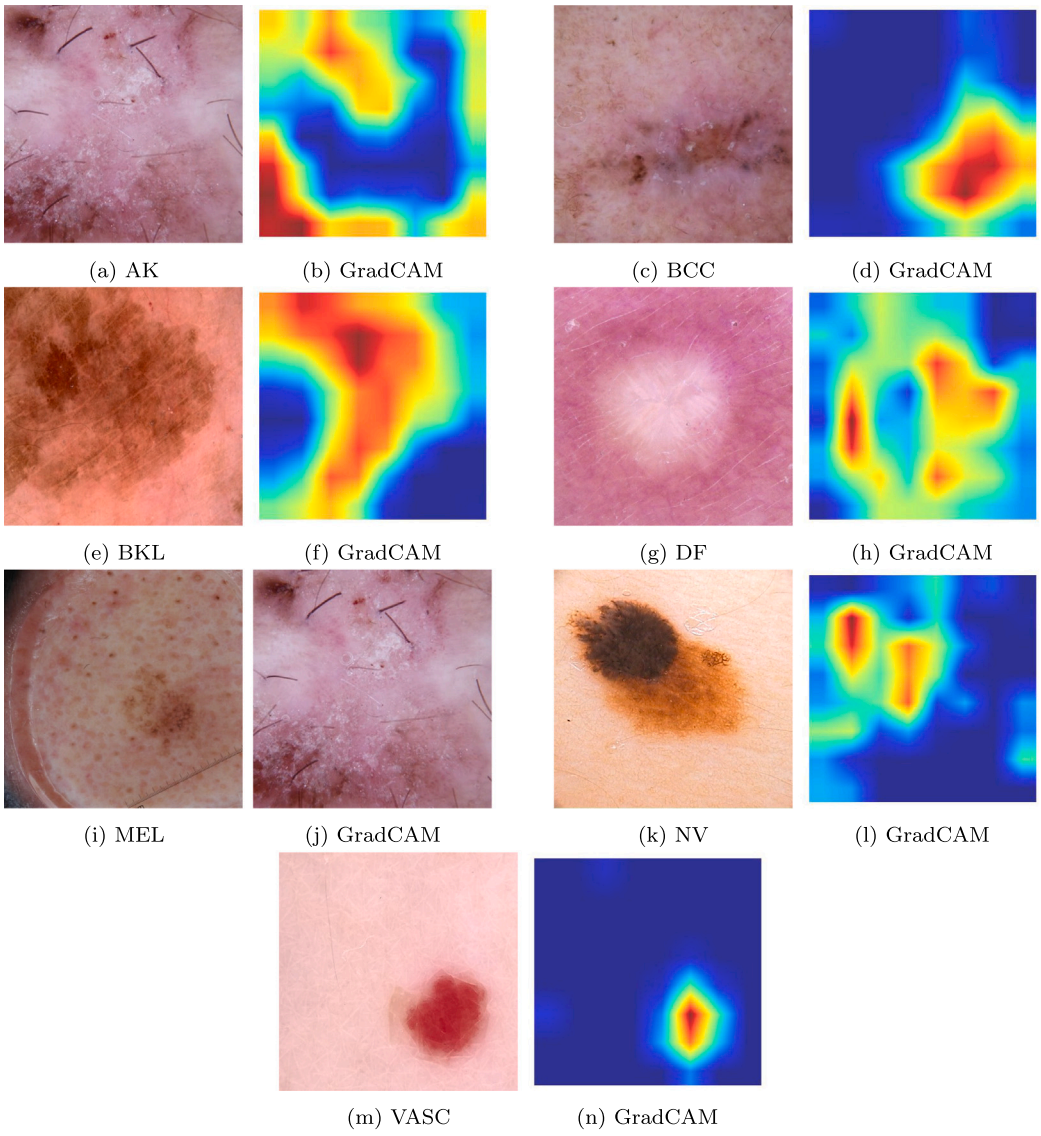


Fig. 14. GradCAM generation by the model for each class.

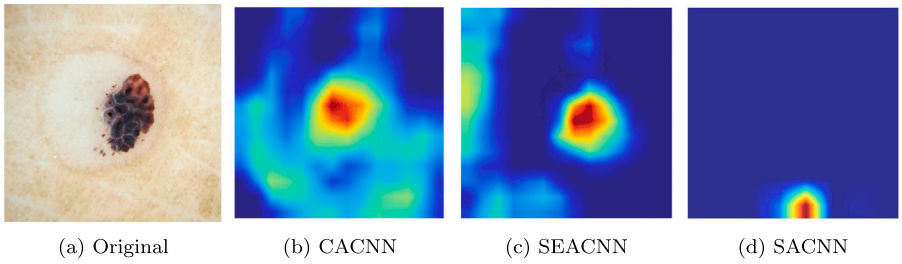


Fig. 15. GradCAM Visualization of Tri-Attention for Architecture Explainability (Example by ENb0 Model)

Specifically, GRA_{F_EN} achieves the highest Accuracy at 94.57% and F1-score at 94.23%, indicating its superior capability in correctly classifying and distinguishing between classes. Additionally, it shows the highest Precision and Specificity, underscoring its effectiveness in minimizing false positives and accurately identifying relevant instances.

In contrast, the models listed without TA, including various configurations of GRA and EN models, exhibit lower performance across all metrics. For instance, the best-performing models among these configurations achieve a maximum Accuracy of 94.08% and an F1-score of 93.75%, which are notably lower than those of GRA_{F_EN}. This performance gap demonstrates the significant advantage of incorporating attention mechanisms into our GRA framework, validating the enhanced performance and effectiveness of GRA_{F_EN} over its TA-less counterparts.

5.6.2. Utilization of conventional ensemble methods instead of GRA

As described earlier, GRA is applied across multiple stages using two different strategies. Predictions from the CCNN, CACNN, SEACNN, and SACNN models are ensembled by computing optimal weights, either across all classifiers or restricted to the top three. When applied at

Table 12
Performances of GRA without TA.

Algorithm	Accuracy	Precision	Recall	F1-Score	Specificity
b0_CCNN	92.13	91.16	92.13	91.35	81.57
b1_CCNN	92.73	92.35	92.73	92.25	85.79
b2_CCNN	92.61	91.75	92.61	92.15	84.18
b3_CCNN	92.73	92.21	92.73	92.21	86.60
b4_CCNN	92.25	91.56	92.25	91.53	83.19
b5_CCNN	92.25	92.31	92.25	92.26	90.37
b6_CCNN	91.89	91.30	91.89	91.34	84.79
b7_CCNN	92.61	91.83	92.61	92.14	86.95
GRA_{ENv1}	94.08	93.84	94.08	93.75	85.30
b0v2_CCNN	91.06	90.40	91.06	90.58	82.77
b1v2_CCNN	92.15	91.63	92.15	91.77	84.25
b2v2_CCNN	91.55	91.03	91.55	91.04	81.37
b3v2_CCNN	91.67	91.29	91.67	91.39	86.17
ENv2S_CCNN	91.55	91.02	91.55	91.05	80.39
ENv2M_CCNN	92.63	92.23	92.63	92.14	84.72
ENv2L_CCNN	92.51	92.19	92.51	92.22	88.60
GRA_{ENv2}	93.72	93.27	93.72	93.33	85.30
GRA_{ENb0}	92.51	91.95	92.51	92.03	82.85
GRA_{ENb1}	92.99	92.68	92.99	92.62	83.80
GRA_{ENb2}	93.24	92.71	93.24	92.72	83.84
GRA_{ENb3}	92.99	92.86	92.99	92.81	88.16
$GRA_{ENb0123}$	94.08	93.75	94.08	93.67	84.84
GRA_{F_EN}	94.57	94.18	94.57	94.23	88.69

stage i, this method is denoted as GRA_i. To demonstrate the effectiveness of GRA, its performance is compared against conventional ensemble techniques, namely Softmax Averaging (SA), Majority Voting (MV), and Weighted Averaging (WA) with random weights. The results of these comparisons are reported below.

Softmax Averaging (SA). Tables 13 summarize the outcomes for Softmax Averaging, where the ensemble applied at stage i is denoted by SA_i. Several configurations of SA achieve competitive performance across different metrics, with $ENv2_{SA_2}$ and $ENb1023_{SA_2}$ emerging as particularly strong performers.

The $ENv2_{SA_2}$ model achieves strong results, with an accuracy of 94.20%, precision of 93.95%, recall of 94.20%, and an F1-score of 93.82%. However, its specificity remains relatively lower at 85.32% compared to GRA_{F_EN}. Similarly, $ENb1023_{SA_2}$ performs competitively, attaining an accuracy of 93.84%, precision of 93.55%, recall of 93.84%, and an F1-score of 93.45%, though its specificity is limited to 84.82%.

In contrast, GRA_{F_EN} demonstrates superior performance across all metrics. It achieves the highest accuracy of 94.57%, surpassing both $ENv2_{SA_2}$ and $ENb1023_{SA_2}$. Its precision (94.18%) and recall (94.57%) also outperform the SA-based models, confirming its stronger detection capability. Furthermore, GRA_{F_EN} secures the highest F1-score of 94.23%, reflecting a more effective balance between precision and recall. Most notably, its specificity reaches 88.69%, substantially higher than both $ENv2_{SA_2}$ and $ENb1023_{SA_2}$.

Overall, these results underscore that GRA_{F_EN} not only surpasses the strongest SA models but also provides a more robust and reliable classification framework.

Majority Voting (MV). As shown in Tables 14, Majority Voting (MV) using all classifiers at stage i is denoted as MV_i.

Several Majority Voting (MV) models achieve strong performance, with $ENv2_{MV_2}$ and $ENv2L_{MV_1}$ standing out as top performers. The $ENv2_{MV_2}$ model records an accuracy of 94.08%, precision of 93.84%, recall of 94.08%, and an F1-score of 93.77%, along with a specificity of 86.75%. Similarly, $ENv2L_{MV_1}$ attains excellent results, achieving 94.44% accuracy, 94.21% precision, 94.44% recall, and an F1-score of 94.27%, with notably high specificity at 91.56%. These findings indicate that both models perform well in terms of classification accuracy as well as their ability to discriminate negative cases.

Table 13
Performance metrics of Softmax Averaging of all classifiers.

Algorithm	Accuracy	Precision	Recall	F1-Score	Specificity
$b0_SA_1$	93.24	92.74	93.24	92.78	84.32
$b1_SA_1$	93.12	92.75	93.12	92.64	83.82
$b2_SA_1$	92.99	92.55	92.99	92.54	85.24
$b3_SA_1$	93.36	93.04	93.36	92.94	84.82
$b4_SA_1$	92.75	92.39	92.75	92.36	85.21
$b5_SA_1$	92.99	92.51	92.99	92.55	85.71
$b6_SA_1$	93.24	92.99	93.24	92.84	84.78
$b7_SA_1$	92.87	92.35	92.87	92.46	85.24
$ENv1_SA_2$	93.48	93.08	93.48	93.07	85.27
$b0v2_SA_1$	92.87	92.39	92.87	92.37	83.83
$b1v2_SA_1$	93.84	93.55	93.84	93.42	86.26
$b2v2_SA_1$	93.12	92.65	93.12	92.62	81.92
$b3v2_SA_1$	93.48	93.01	93.48	93.10	85.29
$ENv2S_SA_1$	93.12	92.66	93.12	92.74	85.74
$ENv2M_SA_1$	92.63	92.06	92.63	92.10	83.77
$ENv2L_SA_1$	94.08	93.75	94.08	93.81	89.63
$ENv2_SA_2$	94.20	93.95	94.20	93.82	85.32
$b0_SA_1$	93.24	92.74	93.24	92.78	84.32
$b0v2_SA_1$	92.87	92.39	92.87	92.37	83.83
SA_ENb0	93.48	93.18	93.48	93.03	84.34
$b1_SA_1$	93.12	92.75	93.12	92.64	83.82
$b1v2_SA_1$	93.84	93.55	93.84	93.42	86.26
SA_ENb1	93.72	93.50	93.72	93.20	83.86
$b2_SA_1$	92.99	92.55	92.99	92.54	85.24
$b2v2_SA_1$	93.12	92.65	93.12	92.62	81.92
SA_ENb2	94.08	93.78	94.08	93.65	84.35
$b3_SA_1$	93.36	93.04	93.36	92.94	84.82
$b3v2_SA_1$	93.48	93.01	93.48	93.10	85.29
SA_ENb3	94.08	93.85	94.08	93.75	86.30
$ENb1023_SA_2$	93.84	93.55	93.84	93.45	84.82
$F_EN_SA_3$	94.08	93.80	94.08	93.69	85.31
GRA_{F_EN}	94.57	94.18	94.57	94.23	88.69

When compared with GRA_{F_EN}, however, the superiority of the latter becomes evident. GRA_{F_EN} delivers the highest accuracy of 94.57%, surpassing both MV models. It also achieves the best precision (94.18%), recall (94.57%), and F1-score (94.23%), reflecting a more effective balance between sensitivity and precision. In terms of specificity, GRA_{F_EN} reaches 88.69%, which, although slightly lower than $ENv2L_{MV_1}$, still exceeds $ENv2_{MV_2}$. Overall, GRA_{F_EN} consistently outperforms the strongest MV ensembles, underscoring its robustness and superior predictive ability.

Weighted Averaging (WA). Tables 15 present the results for Weighted Averaging, where ensembles at stage i are denoted as WA_i. Random weights are assigned to classifiers at each stage as follows: in Stage 1, the four classifiers are weighted 30%, 26%, 24%, and 20% according to descending performance. In Stage 2, the eight ENv1 architectures are combined with weights of 20%, 17%, 14%, 14%, 14%, 10%, 6%, and 5%. For the seven ENv2 models, weights are set to 20%, 18%, 16%, 14%, 12%, 10%, and 10%. Additionally, when combining the common versions of b0, b1, b2, and b3 in Stage 2, weights of 55% and 45% are assigned based on performance. In the final stage, the three predictions are weighted at 40%, 35%, and 25%, respectively.

In the table showcasing the performance of Weighted Averaging (WA) models, $ENv2_{WA_2}$ and $ENv2L_{WA_1}$ emerge as leading performers. The $ENv2_{WA_2}$ model achieves an accuracy of 94.20%, precision of 93.95%, recall of 94.20%, and an F1-score of 93.85%, with a specificity of 86.28%. Similarly, the $ENv2L_{WA_1}$ model demonstrates excellent performance, recording an accuracy of 94.32%, precision of 94.05%, recall of 94.32%, and an F1-score of 94.08%, alongside a high specificity of 90.12%. These results highlight $ENv2L_{WA_1}$ as one of the strongest models within the WA category.

When compared with GRA_{F_EN}, however, the superiority of our approach becomes evident. GRA_{F_EN} achieves the highest accuracy of

Table 14
Performance metrics of Majority Voting of all classifiers.

Algorithm	Accuracy	Precision	Recall	F1-Score	Specificity
$b0_MV_1$	92.99	92.61	92.99	92.67	86.22
$b1_MV_1$	92.75	92.38	92.75	92.48	86.19
$b2_MV_1$	92.63	92.23	92.63	92.34	87.14
$b3_MV_1$	93.36	93.07	93.36	93.06	86.72
$b4_MV_1$	92.63	92.23	92.63	92.26	85.70
$b5_MV_1$	93.12	92.80	93.12	92.85	89.07
$b6_MV_1$	93.00	92.81	93.00	92.73	88.58
$b7_MV_1$	92.99	92.56	92.99	92.68	86.69
$ENv1_MV_2$	93.72	93.37	93.72	93.38	87.20
$b0v2_MV_1$	92.51	92.06	92.51	92.06	83.81
$b1v2_MV_1$	93.48	93.09	93.48	93.13	86.73
$b2v2_MV_1$	93.60	93.15	93.60	93.20	84.82
$b3v2_MV_1$	93.36	93.07	93.36	93.06	86.72
$ENv2S_MV_1$	93.60	93.30	93.60	93.31	86.72
$ENv2M_MV_1$	93.12	92.62	93.12	92.73	86.67
$ENv2L_MV_1$	94.44	94.21	94.44	94.27	91.56
$ENv2_MV_2$	94.08	93.84	94.08	93.77	86.75
$b0_MV_1$	92.99	92.61	92.99	92.67	86.22
$b0v2_MV_1$	92.51	92.06	92.51	92.06	83.81
MV_ENb0	93.12	92.91	93.12	92.94	90.05
$b1_MV_1$	92.75	92.38	92.75	92.48	86.19
$b1v2_MV_1$	93.48	93.09	93.48	93.13	86.73
MV_ENb1	93.12	92.91	93.12	92.94	90.05
$b2_MV_1$	92.63	92.23	92.63	92.34	87.14
$b2v2_MV_1$	93.60	93.15	93.60	93.20	84.82
MV_ENb2	92.51	92.23	92.51	92.32	88.58
$b3_MV_1$	93.36	93.07	93.36	93.06	86.72
$b3v2_MV_1$	93.36	93.07	93.36	93.06	86.72
MV_ENb3	93.36	93.17	93.36	93.20	90.05
$ENb1023_MV_2$	93.12	93.00	93.12	93.01	90.52
$F_EN_MV_3$	93.72	93.47	93.72	93.43	88.16
GRA_{F_EN}	94.57	94.18	94.57	94.23	88.69

Table 15
Performance metrics of Weighted Averaging of all classifiers.

Algorithm	Accuracy	Precision	Recall	F1-Score	Specificity
$b0_WA_1$	93.36	92.88	93.36	92.88	84.33
$b1_WA_1$	92.99	92.73	92.99	92.57	83.81
$b2_WA_1$	93.12	92.63	93.12	92.63	84.77
$b3_WA_1$	93.24	92.88	93.24	92.84	85.28
$b4_WA_1$	92.99	92.62	92.99	92.57	84.75
$b5_WA_1$	92.87	92.39	92.87	92.43	85.70
$b6_WA_1$	93.12	92.82	93.12	92.73	84.78
$b7_WA_1$	92.87	92.29	92.87	92.42	84.77
$ENv1_WA_2$	93.72	93.42	93.72	93.33	84.81
$b0v2_WA_1$	93.24	92.84	93.24	92.76	84.80
$b1v2_WA_1$	93.72	93.39	93.72	93.24	85.31
$b2v2_WA_1$	93.48	93.09	93.48	93.02	82.42
$b3v2_WA_1$	93.24	92.74	93.24	92.81	83.84
$ENv2S_WA_1$	93.48	93.08	93.48	93.11	85.76
$ENv2M_WA_1$	92.51	91.96	92.51	91.99	83.76
$ENv2L_WA_1$	94.32	94.05	94.32	94.08	90.12
$ENv2_WA_2$	94.20	93.95	94.20	93.85	86.28
$b0_WA_1$	93.36	92.88	93.36	92.88	84.33
$b0v2_WA_1$	93.24	92.84	93.24	92.76	84.80
WA_ENb0	93.12	92.71	93.12	92.62	83.84
$b1_WA_1$	92.99	92.73	92.99	92.57	83.81
$b1v2_WA_1$	93.72	93.39	93.72	93.24	85.31
WA_ENb1	93.60	93.37	93.60	93.07	83.38
$b2_WA_1$	93.12	92.63	93.12	92.63	84.77
$b2v2_WA_1$	93.48	93.09	93.48	93.02	82.42
WA_ENb2	94.08	93.75	94.08	93.67	84.36
$b3_WA_1$	93.24	92.88	93.24	92.84	85.28
$b3v2_WA_1$	93.24	92.74	93.24	92.81	83.84
WA_ENb3	93.96	93.73	93.96	93.61	85.81
$ENb1023_WA_2$	93.84	93.55	93.84	93.45	84.82
$F_EN_WA_3$	94.08	93.85	94.08	93.69	85.31
GRA_{F_EN}	94.57	94.18	94.57	94.23	88.69

94.57%, surpassing both $ENv2_{WA_2}$ and $ENv2L_{WA_1}$. It also leads in precision (94.18%), recall (94.57%), and F1-score (94.23%), demonstrating a superior balance between sensitivity and precision. In terms of specificity, GRA_{F_EN} (88.69%) outperforms $ENv2_{WA_2}$ and remains competitive with $ENv2L_{WA_1}$, indicating stronger ability to identify negative cases.

Overall, GRA_{F_EN} consistently surpasses the top WA models across all key performance metrics. Based on these extensive comparisons, our proposed approach, which integrates Tri-Attention (TA) and Gain Ratioed Averaging (GRA), represents an optimal architecture relative to existing state-of-the-art methods.

5.7. Answers to the research questions

Answer to RQ1: To address the variability in image sizes, we develop a scalable architecture using TL employing EfficientNet (EN) models, which inherently support variable image sizes across different versions. By utilizing EN models from all versions, we allow each to train using its respective image size. We then combine these models using the GRA approach, which provides the flexibility to handle different image sizes for classification. This combination leads to the creation of our scalable EfficientNet (SEN).

Answer to RQ2: To emphasize the most significant image features, particularly critical regions, we employ the Tri-Attention (TA) mechanism within a CNN. TA prioritizes these essential areas, ensuring that crucial information is retained as it propagates through the network layers. This approach effectively captures highly relevant details. A key challenge is to balance the risk of neglecting deep features when using a simpler model versus the risk of overfitting with a more complex one. By incorporating TA into the SEN architecture, we achieve superior performance compared to models without any attention mechanism, thereby enhancing overall effectiveness.

Answer to RQ3: Our study demonstrates that employing an ensemble of multiple classifiers is more effective for skin lesion classification than relying on a single model. We design a customized Convolutional Neural Network (CNN) that integrates three distinct attention mechanisms (CA, SEA, and SA), which are subsequently combined with transfer learning (TL) models to construct SEN architectures. Rigorous testing with diverse models and ensemble strategies reveals significant improvements over individual classifiers. Leveraging an ensemble of multiple models enhances both the accuracy and robustness of skin lesion classification by incorporating complementary approaches to feature extraction and decision-making.

Answer to RQ4: Traditional ensemble learning (EL) methods are limited by their inability to determine optimal weights for predictions, particularly when classifier performances vary. Our novel EL approach addresses this by dynamically computing optimal weight ratios for each model. This data-driven strategy improves generalization and enhances performance on unseen data by integrating the most effective prediction ratios. Visual evidence highlights the substantial performance gains of our TS-GRA method, demonstrating improved resilience and efficiency in handling diverse patterns for complex machine learning tasks. The success of this approach is attributed to the innovative use of Gain Ratio, which facilitates the determination of optimal model weights.

6. Discussion and extended comparison

The proposed SEN framework achieves state-of-the-art performance by systematically integrating multiple EfficientNet (EN) variants with advanced CNN architectures. Specifically, our ensemble combines eight

Table 16
Comparison of the proposed SEN + TS-GRA framework against existing approaches.

Article	Accuracy	Precision	Recall	F1-Score	Specificity
[2]	91.51	–	–	–	–
[20]	91.51	–	–	–	–
[18]	89.50	–	89.50	–	**98.10**
[6]	88.00	87.00	94.00	89.00	–
[14]	83.20	–	–	–	–
[16]	90.00	86.00	81.00	86.00	–
[15]	93.40	93.70	–	–	–
[3]	94.00	–	–	–	–
[4]	80.00	–	–	–	–
[19]	86.33	–	86.33	–	97.48
[7]	86.20	91.30	–	–	–
[9]	93.46	–	–	–	92.90
[10]	91.24	83.53	95.04	88.91	–
[11]	86.20	–	–	–	–
[12]	93.46	87.01	85.57	86.28	–
[13]	94.45	93.57	94.01	94.45	–
Ours	**94.57**	**94.18**	**94.57**	**94.23**	88.69

ENv1 variants and seven ENv2 variants with CCNN, CACNN, SEACNN, and SACNN, resulting in a total of sixty distinct models. While each individual model demonstrates strong predictive capability, their collective performance is substantially enhanced through the Gain Ratioed Averaging (GRA) strategy.

The ensembling process unfolds across three hierarchical stages, as defined by our TS-GRA methodology. At each stage, we observe consistent improvements in predictive outcomes, demonstrating the effectiveness of progressively refining model integration. This staged design allows dominant models to retain their influence across levels of ensembling while ensuring balanced contributions from other high-performing models.

By generating predictions from every SEN variant and subsequently refining them via TS-GRA, our method delivers a unified, optimized output. This process highlights the robustness and adaptability of the system, its hierarchical organization, and its precise weighting mechanism, ultimately leading to superior generalization and classification results.

A comprehensive comparison against prior research is presented in Table 16. The results demonstrate that our approach not only surpasses existing methods in terms of all metrics but also achieves competitive specificity. Notably, despite employing a larger pool of base classifiers, the proposed strategy remains computationally feasible while providing clear advantages in predictive reliability and overall performance.

7. Threats to validity

Despite the promising outcomes of our study, several limitations should be acknowledged, as they highlight opportunities for further refinement and exploration:

Large number of classifiers in the initial phase. The inclusion of a broad set of classifiers at the early stage strengthens model robustness but also raises concerns regarding computational overhead. Training and inference with multiple classifiers demand considerable processing power and memory, which may hinder scalability when applied to larger datasets or resource-constrained environments.

Reliance on a single dataset. Our evaluation is based exclusively on one dataset, which restricts the generalizability of the results. This dependence raises questions about how the proposed approach would perform when applied to datasets with different characteristics or levels of complexity. A single-dataset focus may limit the ability of the model to adapt effectively to unseen or heterogeneous data sources.

Recognizing these threats provides direction for future research aimed at improving scalability, efficiency, and the broader applicability of the proposed approach.

8. Conclusion and future work

In this study, we proposed a novel approach for skin lesion classification that integrates customized EfficientNet variants with three complementary attention mechanisms: Channel Attention, Soft Attention, and Squeeze-Excitation Attention. These enhanced models, referred to as *Scalable EfficientNet* (SEN), are further consolidated within a three-stage ensemble learning framework using Gain Ratioed Averaging (GRA) and its hierarchical extension, Triple-Stage GRA (TS-GRA). This combination effectively addresses critical challenges in skin lesion analysis, including class imbalance, optimal feature representation, and adaptive ensemble weighting.

To ensure robust and unbiased evaluation, we applied careful data balancing during training. The attention modules improved feature discrimination, leading to more accurate lesion classification. The proposed GRA method provided an effective post-training mechanism to assign ensemble weights, outperforming conventional strategies and achieving a competitive accuracy of 94.57% on the HAM10000 dataset. Additionally, Gradient-weighted Class Activation Mapping (Grad-CAM) was employed to highlight lesion-specific regions, enhancing interpretability and offering insight into the model's decision-making process.

For future work, we plan to extend this framework to additional dermoscopic datasets, including ISIC, PH2, and PAD UFES 20. We also aim to develop a new dataset by collecting dermoscopic images from hospitals in Bangladesh, facilitating validation in real-world clinical scenarios. Furthermore, we intend to explore methods for reducing the number of base classifiers without sacrificing performance, improving computational efficiency and enabling broader deployment of the SEN + TS-GRA framework.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability statement

The dataset used in this study was sourced from the Harvard Dataverse repository (Tschandl et al., 2018) [31]. After acquisition, the data underwent thorough preprocessing, including the identification and separation of unique and duplicate entries. The unique entries were split into training, validation, and testing sets in a 70:15:15 ratio, respectively, while duplicate entries were included in the training set.

To address class imbalance in the training data, data augmentation techniques were applied, resulting in approximately 8,000 images per class. The augmented datasets were used for model training, and performance evaluation was conducted using the validation and testing sets.

All data, including the augmented training datasets, are publicly available on the Kaggle repository: [HAM10000 Dataset] [37] Author(s) (2024) (https://www.kaggle.com/datasets/ahefatresearch/ham10000-split-and-augmented). The use of the HAM10000 dataset complies with the Creative Commons Attribution-NonCommercial 4.0 International Public License, as we have properly attributed the original dataset and cited the recommended reference paper. Furthermore, our use of the dataset is strictly for non-commercial purposes, aligning with the license's non-commercial clause.

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